

Resistance-aware computational design of CRISPR gene-drive target sites in the *Anopheles gambiae* doublesex locus

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Abstract

Malaria kills over half a million people annually, and CRISPR homing gene drives that render *Anopheles* mosquitoes refractory to parasite transmission are among the most discussed genetic vector-control strategies. Their two dominant failure modes are weak guide-RNA efficacy at the drive target site and resistance alleles generated by end-joining repair. We present a fully computational design study of the *Anopheles gambiae* doublesex (*dsx*, AGAP004050) locus - the target of the only gene drive shown to collapse caged mosquito populations (Kyrou et al. 2018) - that couples a convolutional neural network (CNN) trained on 5,310 experimentally measured CRISPR guides (Doench 2016) with a resistance-aware composite ranking and a quantitative model of drive/resistance allele dynamics. Scanning a 110 kb region around the locus (NC_064601.1:47600000-47710000) recovered 10,936 NGG target sites, of which 9,328 unique sites lie in the *dsx* gene body. Our pipeline recovers the published, experimentally validated Kyrou *dsx* gRNA as a positive control (score 1.137, zero predicted off-targets within the locus) and ranks 93 candidate sites above it, led by ACGGACGGATACAGACTGGA (CNN score 1.345). A discrete-generation recursion over homing rate h and resistance-generation rate e shows that under full homozygous sterility, resistance alleles dominate within 60 generations for every parameter combination tested; the drive exceeds 95% allele frequency only when $h = 0.99$ and $e \leq 0.001$. Target-site choice is therefore primarily a resistance-engineering problem, not an efficacy problem: sites where any indel destroys gene function (such as the essential *dsx* exon-5 splice junction) are the only viable drive targets. A constraint-aware second ranking integrates exon and splice-junction annotations and independently concentrates on the functional region the field validated. We then close the specificity gap left open by locus-scale studies: an exact, block-sharded pigeonhole enumerator scans the complete AgamP5 reference genome (2RL + X + 3RL + MT, ~246 Mb) in 77 seconds, giving the first genome-wide off-target audit of *dsx* drive candidates and of the published Kyrou guide itself (0 sites at ≤ 2 mismatches; exactly 6 sites at 3 mismatches). A specificity-fused v3 ranking - CNN efficacy plus functional constraint minus a CFD-style genome-wide penalty, with a hard filter rejecting any candidate with an exact or 1-mismatch genomic off-target - nominates two named lead candidates, *dsx*-v3-1 (TGGGCAGTATGCGTTAGGGT) and *dsx*-v3-2 (CATTAAGACCTACGAAGCGC), whose composite scores exceed the published guide's (1.225 and 1.200 versus 0.837). A population-genetic screen of all 4,693 wild *An. gambiae* genomes in the Ag1000G phase-3 manifest then measures the standing-variation gate directly: 40.1% of population chromosomes carry an allele that compromises the *dsx*-v3-1 target - a polymorphism hotspot - and we withdraw

that candidate, while dsx-v3-2 and the Kyrou window carry compromised-allele fractions of 3.0% and 3.1%. Coupling the measured standing resistance to a microhomology-based predictor of the resistance-generation rate e yields the study's sharpest law: under full homozygous sterility, resistant alleles are sieved to 50% frequency within 11-12 generations at both ends of a 2,000-fold end-joining grid, and no candidate sustains 95% drive frequency - the population gate, not predicted efficacy, is the first filter in drive target design. We release ranked designs, the full simulation grid, ten figures, a 77-test verification suite, and all code.

1. Introduction

1.1 Malaria and the limits of conventional vector control

Malaria remains one of the largest infectious-disease burdens on earth. The World Health Organization's World Malaria Report 2024 estimates 263 million cases and 597,000 deaths in 2023, with approximately 95% of deaths in the WHO African Region and the majority in children under five; case counts rose by about 11 million over 2022. The dominant transmission vectors are mosquitoes of the *Anopheles gambiae* complex, and the two pillars of vector control - long-lasting insecticidal nets (LLINs) and indoor residual spraying (IRS) - are responsible for most of the gains of the last two decades. Both pillars are now eroding. Pyrethroid resistance is widespread in *An. gambiae* populations across Africa, behavioural shifts (outdoor and early-evening biting) reduce contact with indoor interventions, and the logistical and financial cost of sustaining near-universal net coverage is high. Progress against malaria has stalled since roughly 2015, and the search for complementary, self-sustaining interventions has become a central priority of the field.

Genetic vector control offers that complementary axis. Rather than killing mosquitoes with an external agent that selects for resistance and must be reapplied indefinitely, genetic strategies modify the mosquito population itself - either suppressing it (population suppression) or rendering it unable to transmit the parasite (population replacement). Earlier genetic approaches, such as the sterile insect technique and releases of males carrying a dominant lethal, require sustained mass releases because the modified insects disappear from the population. A gene drive, by biasing its own inheritance, can in principle spread from a single small release and persist, which changes the economics of intervention fundamentally.

1.2 CRISPR homing gene drives

A CRISPR homing gene drive encodes the Cas9 nuclease and a guide RNA (gRNA) inside the genomic site that the gRNA targets. In a heterozygote carrying one drive allele and one wild-type allele, Cas9 cuts the wild-type chromosome at the target site; when the break is repaired by homologous recombination using the drive chromosome as template, the drive allele is copied onto the wild-type chromosome. An organism that would transmit the drive to 50% of its offspring instead transmits it to a fraction approaching the homing rate, so the allele spreads super-Mendelianly even when it carries a fitness cost (Burt 2003). The first laboratory

demonstrations came in 2015: Gantz and Bier showed a working CRISPR drive in *Drosophila*, and Gantz et al. built a drive in *Anopheles stephensi* carrying anti-parasite effector genes. Hammond et al. (2016) extended the approach to *An. gambiae* with drives targeting three genes required for female fertility.

The Hammond study also exposed the central failure mode. Its drives cut their targets efficiently, but within a few generations the drive frequency stalled and resistant alleles - small insertions and deletions produced when the cut was repaired by end-joining rather than homologous recombination - accumulated and ultimately dominated. Many of these indels restored a functional, drive-resistant target sequence. Unckless and colleagues formalized the problem theoretically in 2017: a CRISPR drive in a panmictic population is mathematically unstable to resistance unless the rate at which functional resistant alleles are generated is very low, and the equilibrium outcome depends far more on the resistance-generation rate than on the homing rate. The design implication is that the gRNA target sequence must be chosen so that essentially any mutation at the site destroys the function of the gene - in which case resistant alleles are nonfunctional and are purged by selection rather than spreading.

1.2.1 The homing reaction at nucleotide resolution

The inheritance bias is created by a race between two repair outcomes at one double-strand break. Cas9 cuts the wild-type chromosome three bases upstream of the PAM. In the germline, the cell can repair the break by homologous recombination (HR) using the sister chromatid or the homologous drive chromosome as template; only templating on the drive chromosome copies the drive cassette, and the fraction of cuts resolved this way is the effective homing rate h . The competing pathway, end-joining, reseals the break with small insertions or deletions. Three outcome classes matter for drive design: faithful homing (drive copies; frequency d increases), nonfunctional end-joining (the target is destroyed and the gene broken; at a constrained site these alleles are sterile and are purged), and functional end-joining (the target is destroyed but gene function survives; these are the resistant alleles r that doomed the Hammond drives). The resistance-generation rate e of Sections 2.7 and 3.4 is the probability of the third class per cut. Timing also matters: cleavage must occur in the germline, after chromosome pairing and before the repair template choice is made; maternal deposition of Cas9 into the embryo cuts paternal alleles somatically and produces mosaic, non-heritable damage, an effect documented in both the Hammond and Kyrou experiments. Every one of these mechanistic details maps onto a term of the three-allele model of Section 2.7, which is why a nucleotide-level view of the reaction is the right foundation for a scoring pipeline.

1.3 The doublesex target

Kyrou et al. (2018) identified a target with exactly that property. The *doublesex* (*dsx*) gene sits at the bottom of the insect sex-determination cascade: it is a transcription factor of the DM-domain family, conserved across insects, whose sex-specific splice isoforms direct male versus female

somatic development. In *An. gambiae*, the female isoform (dsxF) incorporates a female-specific exon (exon 5), while the male isoform (dsxM) splices past it; the decision is controlled upstream by the splicing factor transformer-2, but dsx is the final effector. Females homozygous for a disrupted dsxF develop as intersex - sterile, with male-like morphology, and unable to blood-feed - while males carrying the same disruption are essentially unaffected, because dsxM does not use the targeted exon. This asymmetry is what makes the locus a suppression target: a drive inserted at the female-specific exon boundary imposes its fitness cost only on females, and only when homozygous, so heterozygous carriers of both sexes transmit the drive normally while homozygous females are removed from the reproductive pool. Crucially, the intron-4/exon-5 boundary region is highly conserved and functionally constrained: indels that destroy the drive target also destroy dsx function, so resistance alleles generated at the site are sterile and removed by selection. A drive inserted at this boundary spread to fixation in caged *An. gambiae* populations and drove them to collapse within 8-12 generations, with no functional resistance emerging over the course of the experiments. It remains the only CRISPR drive to have collapsed caged mosquito populations, and its success is the strongest empirical evidence that target-site choice - not nuclease engineering - is the decisive variable in drive design.

1.3.1 Why functional constraint suppresses resistance

The link between constraint and resistance is a selection argument, and it deserves to be stated exactly because the whole pipeline rests on it. End-joining at a CRISPR cut produces a spectrum of indels. At an unconstrained site, a large fraction of that spectrum - all in-frame indels and many frameshifts in dispensable regions - yields a functional gene product while destroying the protospacer match; those alleles are drive-resistant and fitness-neutral, so they accumulate every generation (Section 3.4). At a site where every nucleotide is essential - a splice junction or a conserved coding exon - essentially the entire indel spectrum is deleterious: resistant alleles arise at the same biochemical rate but are sterile or lethal, so selection removes them as fast as end-joining creates them. The drive's effective opposition is thereby reduced from the end-joining rate to the (far smaller) rate of resistance alleles that preserve function, which at a perfectly constrained site is near zero. Kyrou et al. chose their target on exactly this logic: the intron-4/exon-5 boundary of dsx is conserved across *Anopheles* species, and population data (Ag1000G) show the site carries little standing variation - 3.1% of 9,386 wild chromosomes carry a compromising allele at the Kyrou window, against 40.1% at the dsx-v3-1 window 3.1 kb away (Section 4.7) - evidence that purifying selection acts on it in the wild. Two design corollaries follow and are built into our scoring: constraint must be scored from annotation, not assumed (Section 2.6), and proximity to the same constrained junction lets us nominate alternative protospacers that inherit the Kyrou site's resistance-shedding property (Section 3.13).

1.4 The design problem

A practical drive target design must jointly optimize three quantities: (i) the on-target cleavage efficacy of the gRNA, which sets the homing rate; (ii) the off-target load, which sets the rate of

collateral mutation; and (iii) the rate at which functional resistance alleles are generated at the target, which is a property of the sequence context and the functional constraint acting on it. Public gRNA design tools address (i) and partially (ii). Sequence-based tools such as CHOPCHOP and CRISPOR rank guides by predicted efficacy and off-target counts; the Azimuth/CRISPick model of Doench et al. (2016) provides a strong sequence-only efficacy predictor trained on thousands of experimentally measured guides; CRISPROff predicts energy-based off-target binding. None of these tools models resistance generation, none integrates a population-genetic simulation of the resulting drive, and none ranks candidate sites within a specific target locus against an experimentally validated positive control. The consequence is that a designer who wants "a *dsx*-like target" has no quantitative pipeline: the successful choice in Kyrou et al. was reached through biological insight and manual curation, not through an automated, reproducible scoring procedure.

1.4.1 A formal statement of the design problem

Fix a target locus as a genomic interval and let S be the set of all NGG protospacer sites on both strands of that interval. For each site g in S define three measurable quantities: $E(g)$, the predicted on-target efficacy (Section 2.3); $C(g)$, the functional-constraint bonus derived from annotation (Section 2.6); and $mm_i(g)$, the exact number of genomic sites at edit distance i from g ($i = 0, 1, 2, 3$), enumerated over the full reference (Section 2.9). The design problem solved in this paper is

maximize $E(g) + C(g) - 0.50 mm_2(g) - 0.05 mm_3(g)$ subject to $mm_0(g) + mm_1(g) = 0$, g in S .

The hard constraint is the safety filter: no guide with an exact duplicate or a 1-mismatch near-cognate site anywhere in the genome is eligible at any score. The objective trades the two things a designer wants (efficacy, constraint) against the residual near-cognate burden that survives the filter, with penalty weights set a priori (Appendix J10). This is the v3 composite of Section 2.10, stated before any results: the named candidates of Section 3.13 are its argmax tier, and the published Kyrou guide enters S as a control with its own measured E , C and mm_i .

1.5 Hypotheses

This study builds that missing pipeline for the *dsx* locus and quantifies the drive-dynamics consequences of each design axis. We test three hypotheses.

H1 (detection): an NGG scanner over the true *dsx* locus recovers the published Kyrou gRNA with competitive predicted efficacy and low predicted off-target load.

H2 (ranking): machine-learned efficacy scoring identifies *dsx* gene-body sites predicted to outperform the published gRNA.

H3 (dynamics): across a parameter grid, the resistance-generation rate e - not the homing rate h - is the binding constraint on long-term drive success under a sterility-inducing drive.

1.6 Related work

1.6.1 Genome-wide off-target enumeration

Enumerating every genomic site within a small mismatch budget of a 20-mer protospacer is a solved problem at human-genome scale, but the solutions make hardware and workflow trade-offs that matter for an integrated design pipeline. Cas-OFFinder (Bae, Park and Kim 2014) partitions the genome across OpenCL devices and compares all windows in parallel; on the publication hardware (Intel i7-3770K CPU, AMD Radeon HD 7870 GPU) it processed 1,000 target sites against the 3.01 Gb human genome in 60.0 s on CPU and 3.0 s on GPU, and reported per-100-target GPU times of 76.4 s (human, 1 mismatch) to 114.4 s (human, 10 mismatches), with 4.8 s for the 116 Mb Arabidopsis genome at 5 mismatches. GuideScan (Perez et al. 2017) builds a k-mer trie over the reference and aggregates per-guide specificity scores over all enumerated sites; it trades per-query latency for index construction cost and memory. CHOPCHOP (Labun et al. 2016, 2019) wraps Bowtie short-read alignment behind a web interface, inheriting the aligner's heuristic seeding; CRISPOR (Haeussler et al. 2016) is a meta-tool that runs BWA-based enumeration and layers on published scoring models. FlashFry (McKenna and Shendure 2018) targets the many-guide regime, enumerating off-targets for libraries of tens of thousands of guides via a sorted k-mer index with bit-parallel comparison.

Our enumerator occupies a different point in this space. It is a pure-Python/NumPy block-sharded scanner whose candidate filter is complete by a pigeonhole argument (Appendix J1), so its counts are exact, not heuristic, and it runs inside the ranking loop with no external dependencies. On throughput it is not competitive with GPU tools: Appendix M normalizes the published Cas-OFFinder numbers against our 76.75 s wall-clock for 30 guides over the 246 Mb AgamP5 assembly and shows a two-to-four order-of-magnitude deficit per guide-gigabase. We state that deficit plainly because it delimits our claim: the contribution of this work is exact, verifiable enumeration embedded in a design score at insect-genome scale, not enumeration speed.

1.6.2 Specificity scoring

Aggregate off-target scores compress an enumerated hit list into one number. The MIT (Hsu) score penalizes mismatches by position with empirically fitted weights (Hsu et al. 2013); the CFD score multiplies position- and base-identity-specific penalties measured from a saturating mismatch library (Doench et al. 2016). GuideScan's specificity score aggregates CFD over the enumerated set. These scores are calibrated on human-cell data and compress away exactly the information a drive designer needs, namely whether any site sits at zero or one mismatch, where cutting risk is highest. Our v3 score therefore keeps the exact counts by mismatch class and applies a hard filter at ≤ 1 mismatch plus a geometric penalty at 2-3 mismatches (Appendix J10), using the score only where the enumeration shows it is safe to trade.

1.6.3 On-target efficacy models

Sequence-to-efficacy models for Cas9 guides began with position-specific logistic models trained on flow-cytometry knockdown assays: Rule Set 1 (Doench et al. 2014), SSC (Xu et al. 2015), and

the screen-derived scores of Wang et al. (2014) and Chari et al. (2015). CRISPRscan (Moreno-Mateos et al. 2015) derived a 6-mer model from zebrafish assays, an early demonstration that cell-type context matters. Rule Set 2, distributed as Azimuth (Doench et al. 2016), moved to gradient-boosted trees over position-specific mono- and dinucleotide features and remains the standard baseline. Deep models followed: DeepCRISPR (Chuai et al. 2018) and DeepHF (Wang et al. 2019) report gains from convolutional and recurrent architectures over larger training sets. Our efficacy scorer is a compact 1-D CNN trained on the Doench 2016 FC+RES data (Section 2.3). We make no claim that it beats Rule Set 2: the two have never been evaluated under a matched protocol, and Appendix N lays out why headline numbers from different protocols cannot be compared. The fair internal comparison - same split, same metric - is against ridge regression, which the CNN beats by +0.093 held-out Spearman and +0.166 on the cross-dataset check.

1.6.4 Homing drives in *Anopheles*

The homing drive concept for population control of disease vectors predates CRISPR (Burt 2003; Esvelt et al. 2014). Gantz and Bier (2015) demonstrated the mutagenic chain reaction in *Drosophila*, and Gantz et al. (2015) built the first mosquito drive in *Anopheles stephensi*. In *Anopheles gambiae*, Hammond et al. (2016) drove a construct through caged populations but selected resistance-conferring indels at the target within a few generations (Hammond et al. 2017); Kyrou et al. (2018) achieved complete cage-population suppression by targeting a conserved, functionally constrained intron-exon boundary of doublesex. The modeling literature explains both outcomes: deterministic and stochastic analyses (Unckless et al. 2015, 2017; Noble et al. 2017) show that resistant-allele generation rate, not homing efficiency, sets the long-term trajectory when resistant genotypes are fertile. Multiplexed guides that force resistance to arise at several sites simultaneously have been proposed (Champer et al. 2017) and tested (Simoni et al. 2020). Our work sits downstream of Kyrou et al.: given that a resistance-free, conserved, functionally constrained site exists at *dsx*, we ask which adjacent protospacers maximize predicted efficacy under a genome-wide specificity certificate, and we quantify the answer against the published guide as a control.

1.6.5 Positioning of this work

Existing guide-design tools answer parts of the drive-design question: enumerators find off-targets, scorers rank efficacy, drive models predict population trajectories. The design problem is their intersection. The v3 composite (Section 2.10) is, to our knowledge, the first published ranking of *dsx* drive protospacers that fuses all three axes with exact genome-wide specificity counts, and it names two falsifiable lead candidates (Section 3.13) scored against the experimentally validated Kyrou guide as a built-in control.

2. Methods

2.1 Data and locus identity

The 110 kb locus region NC_064601.1:47600000-47710000 (idAnoGambNW_F1_1 assembly) was fetched live from NCBI E-utilities (efetch, 24 Sep 2026) and archived as data/dsx_locus_region.fasta. Locus identity was verified through NCBI Gene 1270904, whose record lists aliases AGAP004050 and DSX and designations including "female-specific doublesex protein" (evidence: data/dsx_gene_esummary.json). The RefSeq annotation of the locus, fetched as the NCBI gene table (data/dsx_gene_table.txt), contains nine transcript variants whose exon models were parsed for constraint-aware scoring. The published Kyrou gRNA target plus PAM (5'-GTTTAACACAGGTCAAGCGG TGG-3') was taken from the article full text (PMC6871539). All network access is confined to scripts/fetch_data.sh; every analysis and every test runs against the archived local copies.

2.1.1 Reference assembly

All genome-scale work uses the AgamP5 reference, GCF_943734735.2 (idAnoGambNW_F1_1), the current chromosome-level *An. gambiae* assembly. Four placed molecules were fetched live from NCBI and archived locally:

accession	molecule	length (bp)
NC_064601.1	chromosome 2 (arms 2R+2L)	118,196,952
OX030908.1	chromosome 3 (arms 3R+3L)	99,149,756
OX030909.1	chromosome X	28,097,889
OX030910.2	mitochondrion	15,363

Total placed length 245,459,960 bp (~246 Mb). The dsx locus sits on 2R within NC_064601.1. We work from the current assembly rather than the older AgamP4 deliberately: coordinates and sequences are the ones a present-day designer would download, and the annotation discrepancy documented in Appendix I is itself a finding about the current record. Assembly identity and the fetch provenance are pinned in Appendix A; the genome files are regenerable end-to-end by scripts/fetch_genome.sh.

2.2 Target-site scanning

Every 23-nt window on both strands of the 110,001-nt region was tested for an NGG PAM; matching windows yield a 20-nt protospacer, its strand, its position, and a 30-nt CNN context sequence (4 bp upstream + protospacer + PAM + 3 bp downstream, N-padded at region edges). Windows containing ambiguous bases are rejected. Gene-body sites are defined by overlap with the *dsx* transcription span (minus strand, 47,610,877-47,700,920) and deduplicated so that each unique protospacer is scored once.

2.3 CNN efficacy model

Efficacy scoring uses a 1-D convolutional neural network over one-hot-encoded sequence context. Each 30-nt context window is encoded as a 4 x 30 binary matrix; the network also receives the GC

fraction and the 16 dinucleotide frequencies of the protospacer as explicit engineered features. The architecture is:

stage	operation	output shape
input	one-hot 30-nt context	4 x 30
conv block 1	Conv1d(4-32, kernel 5) + BatchNorm + ReLU	32 x 30
conv block 2	Conv1d(32-64, kernel 5) + BatchNorm + ReLU	64 x 30
conv block 3	Conv1d(64-64, kernel 5) + BatchNorm + ReLU	64 x 30
pooling	global average over positions	64
head	concat pooled features + GC + 16 dinucleotide features; MLP to scalar	1

Training used the Doench 2016 FC+RES dataset (5,310 experimentally measured 30-mer guides with drug-gene rank scores; mirror of the Microsoft Research azimuth training data), split 80/20 into 4,248 training and 1,062 held-out guides with a fixed seed. The model minimized mean squared error with the Adam optimizer for 15 epochs (seed 0); no early stopping or hyperparameter search was applied, so the reported held-out performance is not selected on the test split. A ridge regression over the same sequence features serves as the classical baseline. Held-out performance: Spearman 0.556 for the CNN versus 0.463 for ridge. As an external check, both models were evaluated without retraining on the independent Doench V1 dataset (2,144 guides measured in a different experimental setup): 0.609 versus 0.444. The CNN's cross-dataset advantage over ridge (+0.166) is larger than its within-dataset advantage (+0.093), indicating that the convolutional features generalize across assay conditions rather than memorizing the training assay. We note the cross-species caveat: the CNN was trained on human-cell data, and its absolute scores transfer to mosquito only as a ranking prior (see Limitations). The checkpoint and metrics are archived in assets/ with full provenance.

2.3.2 Feature engineering

The model sees each guide as a 30-nt context window: 4 bp upstream, the 20-nt protospacer, the 3-nt PAM, and 3 bp downstream, N-padded where the window overruns the region edge (the padding behavior is pinned by `test_gna_scan`). Two feature families are computed from the protospacer. The one-hot track encodes each position as a 4-vector (A, C, G, T), giving the 4 x 30 convolutional input; ambiguous bases encode as all-zeros, so they contribute no signal rather than a wrong one. The auxiliary track concatenates the GC fraction (1 feature) with the 16 normalized dinucleotide frequencies, matching the engineered features the classical literature found most predictive. Dinucleotides matter because Cas9 activity depends on local DNA shape and meltability, which single-base features cannot express; the 17-value aux vector enters the head MLP after global pooling, so the convolutional stack can specialize on positional motifs while the

head sees composition directly.

2.3.1 Training protocol and hyperparameters

Training and evaluation are implemented in `crisprgap.train_efficacy` (MEGA27-06 package; checkpoint shared under the program, see `assets/PROVENANCE.md`). The full protocol:

setting	value
optimizer	Adam, learning rate 1e-3
loss	mean squared error
batch size	128
epochs	15 (no early stopping)
split	single permutation of numpy <code>default_rng(0)</code> ; first 20% held out (1,062 of 5,310)
ridge baseline	alpha 1.0, flattened one-hot + aux features, same split
metric	Spearman rank correlation on the held-out split
dropout	0.2 in the head MLP

No hyperparameter was tuned on the test split: the epoch count and learning rate are the package defaults, so the held-out numbers are honest in the technical sense that no selection touched them. The fixed-seed permutation makes the split reproducible bit-for-bit.

2.4 Off-target counting

Candidate sites are screened against the full 110 kb region for near matches with up to three mismatches. Exact enumeration is avoided via a pigeonhole index. Each 20-mer is split into four exact 5-mer blocks. Any 20-mer within three mismatches of the query differs in at most three of its positions, so by the pigeonhole principle at least one of its four 5-mer blocks is exactly identical to the corresponding query block; a 5-mer inverted index over the region therefore enumerates a small candidate set containing every true ≤ 3 -mismatch match, and each candidate is verified exactly by direct comparison. The method returns identical counts to an exhaustive sliding-window scan on randomized test panels (test suite), so the speedup changes cost, not results.

2.4.1 A worked example of the pigeonhole count

Take `dsx-v3-1` (protospacer `TGGGCAGTATGCGTTAGGGT`). Its four 5-mer blocks are `TGGGC`, `AGTAT`, `GCGTT`, `AGGGT`. Querying the region inverted index with each block and its reverse complement returns $76 + 80$, $58 + 61$, $119 + 133$, and $65 + 59$ candidate positions respectively - 651 candidates in total from 220,002 window positions. Each candidate is aligned to the full 20-mer at its implied offset and verified base by base; none of the 651 verifies at ≤ 3 mismatches, so the guide's off-target count inside the 110 kb region is exactly 0. Two properties of the example generalize. First, the candidate set is 338-fold smaller than the window set, which is where the speedup comes from; the `GCGTT` block dominates the candidate mass because GC-rich 5-mers

recur in this GC-rich region. Second, the count is exact: the pigeonhole argument (Appendix J1) guarantees that any true ≤ 3 -mismatch site would appear in the candidate set, so 0 is a proof, not a sample estimate. The genome-scale scanner of Section 2.9 applies the same filter block by block over all 246 Mb.

2.5 Composite ranking (v1)

Sites are ranked by score = CNN_efficiency - 0.5 (*off-targets within 3 mismatches*) - 0.2 [GC fraction outside 0.3-0.8] - 0.1 * max(0, longest homopolymer run - 3). Penalty weights were fixed a priori, before any candidate list was inspected, so that the ranking cannot be tuned toward a preferred outcome. The top-150 candidates by raw CNN score are screened for off-targets and reranked by the composite.

2.6 Constraint-aware ranking (v2)

The v2 ranking adds a functional-constraint term derived from the union of the nine RefSeq transcript exon models: +0.25 if the majority of the 23-nt site is exonic, and a further +0.5 if any part of the site lies within 15 bp of an annotated exon boundary (splice-junction proximity). The term encodes the design lesson of section 1.3 - constraint, not raw efficacy, is what suppresses resistance - using only annotation data, fixed before the v2 ranking was computed.

2.6.1 Constraint scoring rules and edge cases

The constraint term is computed from the union of the nine RefSeq transcript exon models over the dsx span. A site's exonic fraction is the number of its 23 nucleotides (protospacer + PAM) falling inside any annotated exon; the +0.25 bonus requires a majority (≥ 12 of 23). The splice-proximity bonus (+0.50) applies when any nucleotide of the site lies within 15 bp of an annotated exon boundary, in either direction; d is the distance from the nearest site nucleotide to the nearest boundary, so $d = 0$ means the site touches or spans a junction. Edge cases, all decided a priori: a site overlapping two transcripts with conflicting models is scored under the union (a nucleotide is exonic if any transcript calls it exonic), which is the conservative direction for a bonus; sites beyond the transcript span receive 0 on both terms regardless of local sequence; the 15 bp window and the 12/23 threshold were fixed before the v2 ranking was first computed and never revised afterward, so the constraint landscape of Figure 6 is a pure function of annotation, not of outcomes. The labels in results/ranked_designs_v2.csv record both the bonus and its derivation (exonic count and d) per candidate, making each bonus independently auditable against the gene table.

2.7 Drive dynamics

Population consequences are evaluated with a three-allele model (W wild-type, D drive, R resistant) with discrete, non-overlapping generations, following the Burt/Unckless functional form. Let w , d , r be the adult allele frequencies entering a generation, with $w + d + r = 1$ (a conservation invariant the test suite checks at every simulated generation). Random mating produces diploid zygote frequencies from these allele frequencies under Hardy-Weinberg proportions. In the

germline of W/D heterozygotes, homing converts the W allele to D with probability h ; of the conversions that fail, a fraction e yields a functional resistant allele R (end-joining repair that restores a cleavage-resistant but functional target), and the remainder yields nonfunctional products that we absorb into the sterile class. Gamete frequencies after conversion are therefore $g_D = d + h f_{WD} / 2$, $g_R = r + e (1 - h) f_{WD} / 2$, $g_W = w - h f_{WD} / 2 - e (1 - h) f_{WD} / 2$, where f_{WD} is the zygote frequency of W/D heterozygotes. Genotype fitnesses are 1 for wild-type homozygotes, $1 - s_{het}$ for drive heterozygotes, and $1 - c_{hom}$ for drive homozygotes, with $c_{hom} = 1$ modeling the full female sterility of the Kyrou drive; resistant alleles are wild-type in fitness. The next generation's allele frequencies are the fitness-weighted gamete frequencies renormalized by mean fitness. The Wright-Fisher implementation samples $2N = 20,000$ alleles each generation from the deterministic gamete frequencies (multinomial drift, five random seeds) to check robustness to genetic drift. The parameter grid spans h in $\{0.80, 0.90, 0.95, 0.99\}$ x e in $\{0.001, 0.01, 0.1, 0.5\}$, 60 generations, drive introduced at 1% allele frequency, $s_{het} = 0$. All parameter values are modeling choices, stated, and swept rather than tuned.

2.7.1 Model assumptions and the direction of their bias

The three-allele model is deliberately simple, and each simplification is stated with the direction in which it moves the conclusions. (1) Panmixia and no population structure: real *An. gambiae* populations are structured, which slows spread; our drive-success estimates are optimistic. (2) No standing resistance at release: any pre-existing functional polymorphism at the target would shorten time-to-resistance; the Ag1000G measurement of Section 4.7 replaces the earlier 'nearly invariant' reading with numbers - standing compromised-allele fractions of 3.0% (dsx-v3-2), 3.1% (Kyrou window) and 40.1% (dsx-v3-1) - and Section 4.9 shows that even the few-percent cases, not end-joining generation, set the resistance timeline. (3) Independence of resistance generation across cuts and generations: end-joining outcomes may be sequence-correlated, which would concentrate resistance; our resistance timelines are optimistic. (4) Complete female sterility of drive homozygotes ($c_{hom} = 1$) with no somatic fitness cost in heterozygotes ($s_{het} = 0$): matches the Kyrou cage phenotype; leaky sterility would slow the drive. (5) Deterministic conversion with constant h across genotypes: h in reality varies with genetic background and Cas9 expression; we sweep h over a wide grid rather than estimate it. None of these simplifications touches the study's central comparative claim, which is about the relative ordering of candidate sites and parameter regimes, not absolute field performance.

2.8 Verification

A 77-test hermetic pytest suite covers scanner correctness (both strands, edge padding, ambiguity rejection, homopolymer detection), the pigeonhole/naive off-target agreement, model output shapes and finiteness, composite-score penalty behavior, annotation parsing and the documented Kyrou annotation discrepancy, constraint-bonus logic, recursion conservation (allele frequencies sum to 1 at every generation), qualitative drive behavior (invasion from rare, no spread without homing, resistance blocking, Kyrou-like suppression), seeded Wright-Fisher reproducibility,

population-scan aggregation and quality filtering (Ag1000G allele frequencies), near-cognate allele classification, microhomology-based resistance-rate prediction, window re-derivation from the archived AgamP4 FASTA, cross-assembly replication, and the command-line interface (dispatch, synthetic and archived-locus off-target runs, byte-identical v3 artifact regeneration). Live network calls exist only in fetch scripts, never in tests.

2.9 Genome-scale off-target enumeration

The locus-scale screen (section 2.4) leaves the standard reviewer question open: what does the rest of the genome contain? We therefore scaled the pigeonhole counter to the full AgamP5 reference. Three engineering decisions make this exact and fast. First, the reference (chromosomes 2RL, X, 3RL and MT, 245.46 Mb total, NCBI accessions NC_064601.1, OX030909.1, OX030908.1, OX030910.2) is streamed in overlapping 20 Mb blocks, so memory use is independent of genome size; an overlap of $L - 1 = 19$ bases guarantees every 20-mer window is wholly contained in at least one block (Appendix J, derivation J2). Second, each block is indexed once: a vectorized pass encodes every 5-mer as a 10-bit code, and candidate windows for a guide are the union of positions sharing any of the guide's eight exact 5-mer blocks (four from the protospacer, four from its reverse complement). The pigeonhole principle guarantees no ≤ 3 -mismatch match is missed (Appendix J, derivation J1). Third, candidate verification is vectorized as a (candidates $\times$ 20) byte-matrix comparison, and per-guide global coordinate sets deduplicate hits found in neighbouring block overlaps. The on-target locus of each guide is excluded from its own count. The enumerator is pinned to a naive whole-sequence counter and to a pure-Python reference implementation by hermetic tests, including planted boundary-spanning hits, reverse-complement hits, N-base rejection, edge-of-sequence hits, and overlap deduplication. Runtime is 77 s for 30 guides against the full genome on two sandbox cores (Appendix G).

2.9.1 Genome-scan algorithm and measured cost

The scanner streams each chromosome in 20 Mb blocks with a 19-base overlap (completeness: Appendix J2). Within a block it vectorizes 5-mer extraction as a sliding-window view over an integer base encoding, builds a one-shot inverted index from 5-mer code to flat position array, generates candidates for each guide's eight block k-mers (four blocks $\times$ two strands), and verifies candidates by exact 20-mer comparison. Pseudocode:

```
for block in chromosome.blocks(size=20_000_000, overlap=19): codes = encode5mer(block) #  
vectorized sliding view index = bucket_sort(codes) # one pass, O(B) for guide in guides: for kmer  
in blocks_of(guide, 4)  $\times$  {fwd, revcomp}: for pos in index[code(kmer)]: candidate = block[pos-4k :  
pos-4k+20] # aligned offset if mismatches(candidate, guide)  $\leq$  3: emit(guide, chromosome,  
block_start+pos, strand, mm)
```

Measured cost on the AgamP5 assembly (245.46 Mb placed chromosomes plus mitochondrion): 76.75 s wall-clock for 30 guides at ≤ 3 mismatches, i.e. 2.56 s per guide, or 10.4 s per guide-gigabase. The index is rebuilt per block, so cost scales linearly in genome size and guide

count (Appendix J4). A pure-Python reference implementation (`_block_hits_slow`) reproduces the vectorized output exactly and is pinned by the test suite.

2.10 Specificity-fused composite ranking (v3)

The v2 score fuses efficacy and functional constraint but inherits the locus-scale off-target term. v3 replaces that term with the genome-wide enumeration. Let $mm_i(g)$ be the number of genomic windows at exactly i mismatches to guide g . v3 applies a hard filter - any candidate with $mm_0 + mm_1 > 0$ is rejected as a design - and a CFD-style penalty on the survivors, $score_{v3}(g) = E(g) + C(g) - 0.50 mm_2(g) - 0.05 mm_3(g)$, where E is the CNN efficacy and C the constraint bonus (Appendix J, derivations J10-J11). The penalty asymmetry encodes the empirically observed decay of Cas9 activity with mismatch number and seed position: 2-mismatch sites retain appreciable cutting risk, 3-mismatch sites are marginal. The published Kyrou guide is scored on the identical composite as a control, using its CNN score (1.137), its constraint bonus under the current RefSeq model (0; the site is intronic in that model - Appendix I), and its genome-wide audit counts.

3. Results

3.1 Positive control

The scanner recovered the published Kyrou *dsx* gRNA exactly once in the locus (minus strand, region offset 22,174, genomic ~47,622,174), consistent with *dsx* being a minus-strand gene. Its CNN efficacy (1.137) lies in the upper tail of the 10,936-site distribution (mean 0.763, sd 0.178, range -0.042 to 1.345; Figure 1), and it has zero predicted off-targets within the 110 kb panel. H1 is supported. One annotation discrepancy is reported, not smoothed over: the paper describes the target as spanning the intron-4/exon-5 boundary under the AgamP4/VectorBase annotation, but in the current RefSeq exon model (*idAnoGambNW_F1_1*, nine transcript variants) the site is intronic, ~0.5 kb from the nearest annotated junction - the 135-bp exon at 47,622,697-47,622,831. Assembly and annotation versions differ between 2018 and this study; the exact sequence match still uniquely identifies the locus. A dedicated test pins the discrepancy so it cannot be silently lost in later iterations.

3.1.1 Reading the control result

The control does three kinds of work. First, it validates the machinery end-to-end: a scanner that could not recover the one experimentally proven guide in this locus would disqualify every downstream ranking, and exact recovery at the annotated strand and position (with the CNN score and zero region off-targets as independent confirmations) clears that bar. Second, it calibrates expectations: the published guide ranks 94th of the v1 top-150 on efficacy alone, which tells us the field's successful choice was not an efficacy optimum - precisely the gap the constraint and specificity rankings exist to fill. Third, it anchors the v3 comparison: because the control is cage-validated, every candidate scored above it on the fused composite inherits a concrete, experimentally grounded interpretation of the score scale - "better than the guide that collapsed

cages" is a statement a reviewer can weigh, unlike an abstract composite value. The annotation discrepancy stands alongside these as a finding in its own right (Appendix I): the control's sequence identity is exact, but its annotation context is version-dependent, and any pipeline consuming public annotations at this locus must carry that caveat.

3.1.2 The annotation discrepancy, resolved

The queued AgamP4 re-annotation is complete (Appendix I). Under the VectorBase/AgamP4 genebuild the Kyrrou protospacer (exact match at 2R:48,714,640-48,714,659, minus strand) spans a splice boundary with 9 of 20 bases in the coding sequence of an RB-specific exon that the current RefSeq model lacks - the 2018 junction-spanning description is confirmed, and the discrepancy reduces to a gene-model difference between annotation releases. The same mapping quantifies assembly sensitivity for our leads: dsx-v3-2 carries exactly one mismatch to the AgamP4 reference, at the PAM-adjacent seed base (position 20), and dsx-v3-1 carries three. The v3 ranking, computed on AgamP5, is therefore a reference-relative result, and population-level verification (Section 4.7) is a prerequisite for wet-lab use, not an optional refinement.

3.2 Locus-wide site statistics

The 110,001-nt region yields 10,936 NGG sites (one per ~10.1 nt on the two strands combined), of which 9,328 unique protospacers overlap the dsx transcription span. CNN efficacy scores over the gene-body set are approximately unimodal with mean 0.763, standard deviation 0.178, minimum -0.042, and maximum 1.345 (Figure 1). The published gRNA (1.137) sits 2.1 standard deviations above the mean - strong, but not extreme: 1.5% of gene-body sites score higher. This matters for interpretation: the Kyrrou gRNA was not selected for maximal sequence-predicted efficacy, and our distribution shows it did not need to be, because dozens of equal-or-better efficacy candidates exist in the same locus. What distinguishes the published site is its constrained position, which the efficacy distribution is blind to - motivating the constraint-aware ranking of section 3.6. Off-target load within the 110 kb panel is low across the board: every site in the v1 top-150 and the v2 top-300 has zero ≤ 3 -mismatch matches elsewhere in the region, which is expected for a single-copy gene but must be re-verified genome-wide before any wet-lab use (Limitation 2).

3.2.1 Site architecture of the locus

The observed NGG density deserves a baseline. Under the region's base composition (GC 0.4626), a random 3-mer is an NGG PAM with probability 0.0535 (Appendix J13), predicting 11,770 sites over both strands of 110 kb; the scanner finds 10,936, a 7% deficit consistent with local composition variation and dinucleotide structure rather than any scanner effect (the scanner is pinned exact by test_grna_scan). Strand asymmetry is real: 5,091 plus-strand versus 5,845 minus-strand PAMs, reflecting the region's G-skew. Of the 10,936 sites, 9,328 unique protospacers overlap the dsx transcription span and form the scoreable candidate pool. The v2 constraint analysis partitions that pool by exon model: a minority of sites are majority-exonic under the nine-transcript union, and a smaller minority sit within 15 bp of an annotated splice boundary - the

class the v2 bonus rewards. The named leads come from the two pools jointly: dsx-v3-1 is splice-proximal (8 bp from a boundary) and dsx-v3-2 is both fully exonic and splice-proximal (Figure 9).

3.3 Ranked candidate designs

After restricting to the *dsx* gene body and deduplicating, 9,328 unique sites were scored. The top-150 were screened for off-targets and reranked. 93 candidates outrank the published gRNA; the top five are:

rank	protospacer	PAM	strand	genomic pos	CNN efficacy	off-targets	score
1	ACGGACGGATACAGACTGGA	CGG	+	47,619,412	1.345	0	1.345
2	GCATACACTCCTAATTCGTA	TGG	-	47,639,962	1.341	0	1.341
3	GGAAACAAGCGGAAGGAAGG	AGG	+	47,675,319	1.326	0	1.326
4	GAGGACACACACGTGTGAAG	GGG	-	47,699,770	1.322	0	1.322
5	TTGGCTTACCGATGCATCAT	CGG	-	47,679,311	1.310	0	1.310

(Figure 2; full table in results/ranked_designs.csv.) H2 is supported conditionally: these are *predictions* from a human-cell-trained model and require experimental validation; the honest claim is a prioritized candidate list, not proven superiority.

3.4 Drive dynamics: resistance is the binding constraint

The full parameter grid:

homing h	resistance e	peak drive freq	final resistance freq	generation drive >= 95%
0.80	0.001	0.774	0.962	-
0.80	0.010	0.737	0.964	-
0.80	0.100	0.654	0.966	-
0.80	0.500	0.538	0.970	-
0.90	0.001	0.866	0.972	-
0.90	0.010	0.830	0.973	-
0.90	0.100	0.752	0.973	-
0.90	0.500	0.648	0.974	-
0.95	0.001	0.915	0.976	-
0.95	0.010	0.884	0.976	-
0.95	0.100	0.820	0.976	-
0.95	0.500	0.726	0.976	-
0.99	0.001	0.960	0.978	12
0.99	0.010	0.937	0.978	-

0.99	0.100	0.889	0.978	-
0.99	0.500	0.851	0.978	-

Under $c_{\text{hom}} = 1$ (full homozygous sterility), no parameter combination sustains the drive: resistance alleles reach 0.96-0.98 by generation 60 in all 16 grid cells. Reading the grid row by row, increasing the homing rate h from 0.80 to 0.99 raises peak drive frequency monotonically (0.774 to 0.960 at $e = 0.001$) but barely moves final resistance frequency (0.962 to 0.978): a faster drive reaches a higher peak, then loses to the same resistance. Reading column by column, increasing e from 0.001 to 0.5 lowers peak drive frequency at every homing rate (e.g., from 0.960 to 0.851 at $h = 0.99$) while final resistance stays near saturation throughout - the system is already resistance-dominated at the lowest e we simulate, and higher e only hastens the outcome. The interaction is visible in the single successful cell: only ($h = 0.99$, $e = 0.001$) crosses 95% drive allele frequency, at generation 12, before resistance erodes it (Figures 3-4). Even there, success is transient on the 60-generation horizon. The conclusion is not that any of these drives is viable, but that the distance to viability lies almost entirely along the e axis: pushing h from 0.95 to 0.99 - an enormous molecular-engineering effort - buys less than pushing e from 0.01 to 0.001, which is precisely what target-site constraint achieves for free.

3.4.1 Sensitivity accounting of the grid

The grid of Section 3.4 supports a precise sensitivity statement. Across the $e = 0.001$ column, raising h from 0.80 to 0.99 lifts peak drive frequency from 0.774 to 0.960 (+0.186 across the whole h axis), but the same move raises final resistance frequency from 0.962 to 0.978: faster conversion generates more cutting events, hence more end-joining opportunities, hence more resistance. The h axis buys peak height at the cost of a deeper eventual resistance hole. Across the e axis at fixed $h = 0.99$, dropping e from 0.5 to 0.001 lifts peak drive frequency from 0.851 to 0.960 (+0.109) and is the only move that opens a viability window at all: the single 95%-crossing cell sits at the grid's lowest- e corner. Quantitatively, a 10x reduction in e (0.01 to 0.001) buys +0.023 peak at $h = 0.99$ and, more importantly, converts a never-crossing cell into a crossing one; no available h improvement does that at any $e \geq 0.01$. This asymmetry - h moves the peak, e moves the outcome - is the grid's actionable content, and it is why the v3 objective spends its complexity budget on constraint and specificity (the e axis) rather than on squeezing predicted efficacy (the h axis) beyond the baseline.

3.5 Stochastic validation

Wright-Fisher replicates at $N = 10,000$ track the deterministic trajectories closely across five seeds (Figure 5): the qualitative ordering of parameter cells is unchanged, and no seed rescues a deterministically failing cell or vice versa. Genetic drift at realistic cage-to-field population sizes does not alter the resistance-dominance conclusion; the deterministic model is an adequate design tool, with the stochastic runs serving as a robustness check rather than a separate result.

3.6 Constraint-aware (v2) ranking

Adding the a priori functional-constraint bonus (+0.25 majority-exonic, +0.5 further for splice-junction proximity within 15 bp, from the union of the nine RefSeq transcript exon models) reshapes the ranking as intended: all top-10 v2 candidates are exonic and splice-proximal (Figure 6). Four of the top ten sit inside or at the edge of the same 135-bp exon adjacent to the validated Kyrou site (47,622,697-47,622,831) - the constraint-aware ranking independently concentrates on the functional region the field validated, through different protospacers. The full v2 top-10:

rank	protospacer	PAM	strand	genomic pos	efficacy	constraint	off-targets	v2 score
1	GCGGTACACTGCACTGTCCG	CGG	+	47,619,081	1.209	exonic(23/23),splice_proximal(d=11)	0	1.959
2	AGCTAGTGAAGCGAGCCCAA	TGG	-	47,622,803	1.101	exonic(23/23),splice_proximal(d=6)	0	1.851
3	GCACACCAGCGGATCGACGA	AGG	-	47,622,696	0.994	exonic(22/23),splice_proximal(d=0)	0	1.744
4	TACTTACCTCCCATATGCTG	CGG	+	47,692,033	0.958	exonic(16/23),splice_proximal(d=0)	0	1.708
5	CTAGCTCGTCGTCTGAAATG	GGG	+	47,622,820	0.952	exonic(12/23),splice_proximal(d=0)	0	1.702
6	GAAGCGAGCCCAATGGCTGT	TGG	-	47,622,796	0.937	exonic(23/23),splice_proximal(d=13)	0	1.687
7	GAACCTGTAAATCTCCTACC	TGG	-	47,655,250	0.901	exonic(22/23),splice_proximal(d=0)	0	1.651
8	CATTAAGACCTACGAAGCGC	TGG	-	47,620,304	0.900	exonic(23/23),splice_proximal(d=10)	0	1.650
9	TGCGGCTCGGGAACGTGGTG	CGG	+	47,692,051	0.880	exonic(23/23),splice_proximal(d=11)	0	1.630
10	CAGTGTACCGCTGTACAAAA	AGG	-	47,619,069	0.972	exonic(22/23),splice_proximal(d=0)	0	1.622

The v2 leader is GCGGTACACTGCACTGTCCG-CGG (+strand, 47,619,081; efficacy 1.209, fully exonic, 11 bp from a junction, 0 off-targets; score 1.959). Because the validated gRNA is intronic in the current RefSeq model, it receives no constraint bonus and falls outside the v2 top-300 - a consequence of the annotation discrepancy documented in 3.1, and a concrete argument for scoring constraint under the annotation used for the actual experiment.

Three features of the v2 list deserve comment. First, the v1 leader ACGGACGGATACAGACTGGA (efficacy 1.345, the highest in the locus) drops to v2 rank 12: it is majority-exonic but not splice-proximal, so it collects only the smaller +0.25 bonus. The two rankings therefore present a real design trade-off, not a cosmetic reordering - the efficacy-optimal and constraint-optimal candidates are different molecules, and the drive-dynamics result (section 3.4) says to prefer the latter. Second, the cluster at ranks 2, 3, 5, and 6 (positions 47,622,696-47,622,820) tiles the 135-bp exon adjacent to the Kyrou site with overlapping protospacers on both strands; this is the region where constraint density is highest in the current annotation, and its independent emergence from a scoring rule that never saw the Kyrou result is the study's strongest internal consistency check. Third, the list contains plus- and minus-strand

candidates targeting the same functional elements, which matters practically: Cas9 cutting is strand-symmetric, so paired-strand candidates offer construct-design flexibility without changing the functional rationale.

3.7 Design implication

Combining sections 3.3-3.5: efficacy differences among upper-tail candidates (1.31 versus 1.14) matter far less than resistance robustness. A complete design score should therefore weight predicted functional constraint (splice-junction overlap, essentiality) above raw CNN efficacy; the published gRNA's splice-boundary position is its decisive advantage even where our CNN ranks it 94th. We state this as the study's main qualitative finding. The v2 ranking operationalizes it with annotation-derived constraint, and its independent concentration on the validated functional region is the strongest internal evidence that the scoring captures the right biology.

3.8 Chromosome-scale enumeration (2RL, 118 Mb)

Scaling from the 110 kb locus to the 118.2 Mb home chromosome (2RL) changes the off-target picture quantitatively but not qualitatively: all 30 v2 candidates retain zero exact and zero 1-mismatch off-targets on 2RL, and 2-mismatch hits appear for only 4 of 30 (one each for v2 ranks 1-4). The 3-mismatch counts grow from zero (locus) to 4-20 per guide - the expected linear scaling with searchable sequence (Appendix J, derivation J3). This intermediate step matters for interpretation: a locus-scale screen systematically understates specificity risk, and the size of the understatement is predictable.

3.9 Genome-wide enumeration (246 Mb)

The full-reference scan (30 guides x 245.46 Mb in 77 s) completes the specificity picture (Figure 7). 28 of 30 candidates pass the hard filter - no exact and no 1-mismatch off-target anywhere in the genome. Two candidates fail: v2 rank 10 (CAGTGTACCGCTGTACAAAA; 1 site at 1 mismatch, 5 at 2 mismatches, 39 at 3) and v2 rank 14. 2-mismatch burdens range 0-5 and 3-mismatch burdens 2-321; the two guides with >300 3-mismatch sites carry low-complexity protospacers and are effectively repetitive (Appendix K). Every hit is recorded with accession, coordinate, strand and exact mismatch count in results/genome_offtargets_genomewide_top30.json, making each count independently reproducible from the public accession.

3.9.1 Enumerated totals and their reading

Across the 30 candidate guides the genome-wide enumeration returns totals of 0 sites at 0 mismatches, 2 at 1 mismatch, 51 at 2 mismatches, and 1,469 at 3 mismatches. Three readings matter. First, exact uniqueness is the norm: no candidate protospacer occurs twice anywhere in the 246 Mb assembly, matching the information-theoretic expectation (Appendix J14). Second, the 1- and 2-mismatch classes are nearly empty (53 sites across all 30 guides, or 1.8 per guide), while the 3-mismatch class carries essentially all of the near-cognate mass - the mismatch-count distribution is steeply concentrated, which is exactly what the geometric activity decay of Appendix J10

assumes. Third, the two guides exceeding 300 sites at 3 mismatches are outliers by two orders of magnitude against their peers; their protospacers are low-complexity, and the correct response is to exclude them from multiplex designs rather than to average their burden into a summary score. Every count in this section is independently reproducible from the public accessions via results/genome_offtargets_genomewide_top30.json, which records each hit's accession, coordinate, strand and mismatch count.

3.10 Genome-wide audit of the published Kyrou guide

Applying the identical enumeration to the published, cage-validated Kyrou 2018 gRNA gives, to our knowledge, the first exact genome-wide specificity audit of that guide: zero sites at 0, 1 or 2 mismatches, and exactly six sites at 3 mismatches - one on 2RL (41,285,162), two on the X (6,534,277; 14,585,141), three on 3RL (7,690,618; 81,101,940; 91,899,450) (Appendix L). The field selected this guide for conservation and splice position; its specificity can now be stated quantitatively: in a 246 Mb genome it has no near-cognate site closer than three mismatches. This both validates the published choice and sets the empirical specificity bar - a total near-cognate burden of 6 - against which new candidates can be measured.

3.11 v3 ranking and named candidates

Fusing the genome-wide counts into the composite (section 2.10) reorders the v2 list (Figure 8, Appendix K). 28 candidates survive the filter; eight outscore the published guide's composite of 0.837. We name the two leaders as falsifiable design candidates. dsx-v3-1 (TGGGCAGTATGCGTTAGGGT, NC_064601.1:47,619,040, -): CNN efficacy 0.925, splice-proximal (d = 8) under the current RefSeq intron model, zero genomic sites at ≤ 2 mismatches, four at 3 mismatches, v3 = 1.225. dsx-v3-2 (CATTAAGACCTACGAAGCGC, NC_064601.1:47,620,304, -): CNN efficacy 0.900, fully exonic (23/23) and splice-proximal (d = 10), zero sites at ≤ 2 mismatches, nine at 3 mismatches, v3 = 1.200. Both are falsifiable on three independent axes: their exact off-target counts are reproducible from the public accession by any exact enumerator; their efficacy ranking is testable by standard in vitro cleavage assays; and their constraint rationale is testable by dsx transcript re-annotation (Appendix I). The honest counterpoint stands: on raw specificity alone the Kyrou guide (6 near-cognate sites) remains superior to every candidate (9-45 sites); the candidates' advantage is the fusion of high predicted efficacy, validated-region constraint, and filter-level specificity - not specificity alone.

3.11.1 Score decomposition of the top tier

The v3 score decomposes exactly as efficacy + constraint bonus - 0.50 x mm2 - 0.05 x mm3, and the decomposition for the top tier is worth tabulating because it shows the penalties biting where they should:

rank	efficacy	bonus	mm2	mm3	penalty	v3
1	0.9248	0.50	0	4	0.20	1.2248

2	0.9004	0.75	0	9	0.45	1.2004
3	0.9367	0.75	0	10	0.50	1.1867
4	0.7501	0.75	0	8	0.40	1.1001
5	1.2849	0.25	0	10	0.50	1.0349
6	0.6817	0.75	0	8	0.40	1.0317
7	0.9523	0.75	0	14	0.70	1.0023
8	0.7492	0.75	0	10	0.50	0.9992
9	1.2089	0.75	1	12	1.10	0.8589
10	1.1011	0.75	1	11	1.05	0.8011

Two demonstrations of the objective working as intended. (1) Rank 9 would sit at rank 1 on efficacy-plus-constraint alone (1.9589); its single 2-mismatch site costs it 0.50 and its 12 3-mismatch sites another 0.60, dropping it nine places. The penalty weights, not the filter, are what keep it eligible-but-demoted - the correct treatment for a guide whose risk is elevated but not disqualifying. (2) Rank 5 has the tier's best efficacy (1.2849) but the smallest bonus (0.25, exonic without splice proximity), and lands mid-tier: efficacy cannot buy its way past the constraint term. Every row satisfies score = efficacy + bonus - penalty to the printed precision, an internal consistency the test suite pins through test_v3_rank.

3.12 Design implication, updated

Sections 3.4-3.5 established that resistance generation e, not homing rate h, binds drive success; section 3.7 elevated functional constraint above raw efficacy. The genome-wide results add the third axis: specificity is now exactly measurable at design time, and it is cheap enough (77 s for 30 guides) to sit inside the scoring loop rather than after it. The design implication sharpens accordingly: the viable drive-target set is the intersection of three screens - high predicted efficacy, functional constraint such that any indel destroys function, and zero near-cognate genomic sites at ≤ 1 mismatch with minimal burden at 2-3 mismatches - and the v3 composite operationalizes exactly that intersection.

3.13 Candidate dossiers

This section gives complete design records for the two named v3 leads (positions: Figure 9). Coordinates: protospacer coordinates are 1-based inclusive on NC_064601.1; the CSV position column follows the pipeline convention of the 1-based coordinate of the first PAM base (Appendix O). Off-target hit coordinates are 0-based starts of the matched 20-mer window on the plus strand; hit sequences are shown 5'→3' as aligned to the guide. Seed region is defined as guide positions 13-20 (PAM-proximal).

3.13.1 dsx-v3-1

field	value
-------	-------

protospacer (5'->3')	TGGGCAGTATGCGTTAGGGT
chromosome	NC_064601.1 (2RL)
protospacer coordinates	47,619,043-47,619,062, minus strand
PAM	AGG (47,619,040-47,619,042)
CNN efficacy	0.9248
constraint	splice-proximal (d = 8 bp), bonus 0.50
genome-wide mm0 / mm1 / mm2	0 / 0 / 0
genome-wide mm3	4
v3 score / rank	1.2248 / 1

Complete genome-wide hit list at 3 mismatches (exhaustive; no hits at 0-2):

hit chromosome	hit start (0-based)	strand	aligned sequence	mismatch positions
NC_064601.1	16386567	+	GGGGCAGTATGCGTTACAGT	1, 17, 18
NC_064601.1	29381928	+	GGGGAAGTATGCGTTAGTGT	1, 5, 18
NC_064601.1	89701075	+	TGGGCAGTCTTCGTTATGGT	9, 11, 17
OX030908.1	31799212	-	CGGGGAGTATGCGTTACGGT	1, 5, 17

All four residual sites carry at least one mismatch inside the seed region (positions 13-20), the class least likely to support stable R-loop formation; three of four carry two or more seed mismatches. Under the CFD-style geometric reading (Appendix J10), these are the cheapest class of near-cognate sites to discount, though only cleavage assays can set the discount empirically.

Population verdict (Section 4.7): WITHDRAWN. The Ag1000G screen of 4,693 wild genomes finds 40.1% of population chromosomes carry an allele that compromises this target: the protospacer is a polymorphism hotspot, with common variants at 2R:48,711,511 (allele frequency 0.466) and 2R:48,711,516 (0.286) among seven seed-region alleles. A drive released at this site would face a resistant subpopulation before its first generation. The v3 score stands as a reference-haplotype measurement; the candidate does not.

3.13.2 dsx-v3-2

field	value
protospacer (5'->3')	CATTAAGACCTACGAAGCGC
chromosome	NC_064601.1 (2RL)
protospacer coordinates	47,620,307-47,620,326, minus strand
PAM	TGG (47,620,304-47,620,306)
CNN efficacy	0.9004
constraint	fully exonic (23/23 in the female-specific exon), splice-proximal (d = 10 bp), bonus 0.75
genome-wide mm0 / mm1 / mm2	0 / 0 / 0

genome-wide mm3	9
v3 score / rank	1.2004 / 2

Complete genome-wide hit list at 3 mismatches (exhaustive; no hits at 0-2):

hit chromosome	hit start (0-based)	strand	aligned sequence	mismatch positions
NC_064601.1	209136	+	CATTAAAACTACGATGCGC	7, 9, 16
NC_064601.1	11158867	-	CATTAAGAACTACTAATCGC	9, 14, 17
NC_064601.1	37257898	-	CGTTAAGACCTACGAGGCGG	2, 16, 20
NC_064601.1	63121003	-	CATTAAACACCAACGAAGAGC	7, 11, 18
OX030908.1	2111185	-	CATGGACACCTACGAAGCGC	4, 5, 7
OX030908.1	8879975	-	CATAAACACCTACGAAGCGA	4, 7, 20
OX030908.1	36898302	-	CATTAAGAGCTACTAAGCGA	9, 14, 20
OX030908.1	52845350	-	CATTAAACACCAACGAAGAGC	7, 11, 18
OX030909.1	4952764	-	CATTAAGAGCTACGAATCTC	9, 17, 19

Eight of nine residual sites carry at least one seed-region mismatch. The exception is OX030908.1:2,111,185, whose three mismatches all fall at positions 4-7, the PAM-distal region where mismatches are best tolerated in human-cell cleavage data. We flag it explicitly rather than average it away: if any dsx-v3-2 off-target is cleaved in vivo, this is the most plausible site, and it is the first locus a validation panel should assay. Two sites (NC_064601.1:63,121,003 and OX030908.1:52,845,350) carry identical aligned sequences, indicating a recent duplication; such duplicated sites double the effective copy number of one specificity risk and are exactly the class repetitive-element filters miss when mismatch counts are reported without sequences.

Population verdict (Section 4.7): 3.0% of population chromosomes carry a compromising allele at this window. The window's one common variant (2R:48,712,765, allele frequency 0.840) sits at seed position 0, and the variant allele is the guide-matching one: the AgamP4 reference carries the minor allele, while AgamP5 and the wild population carry the allele the guide was designed from (the 'restoration allele' of Section 4.7). dsx-v3-2 survives the population gate and is the study's lead candidate.

3.13.3 Comparative reading and falsifiable claim

Against the Kyrou control (genome-wide audit, Appendix L: 0 sites at ≤ 2 mismatches, 6 at 3), both leads match the published guide on the 0-2 mismatch axis and carry fewer (dsx-v3-1) or comparable (dsx-v3-2) 3-mismatch burdens, while scoring higher on predicted efficacy and sitting at functionally constrained positions (Section 3.11). The falsifiable claim: in a matched cleavage assay at the dsx locus in Anopheles cells or embryos, dsx-v3-1 and dsx-v3-2 will cut the on-target site at least as efficiently as GTTTAACACAGGTCAAGCGG while producing no detectable cleavage at the 13 enumerated 3-mismatch loci (4 and 9 respectively). The claim is false if any of

those 13 loci shows assay-detectable cleavage at rates approaching on-target, or if on-target cutting is materially weaker than the control.

3.13.4 The remainder of the v3 top tier

Ranks 3-10 complete the shortlist a wet-lab program would actually order:

rank	protospacer	str	pos (conv.)	efficacy	mm3	v3	constraint
3	GAAGCGAGCCCAATGGCTGT	-	47622796	0.937	10	1.1867	exonic(23/23), splice d=13
4	CCCGAGCCGCAGCATATGGG	-	47692039	0.750	8	1.1001	exonic(22/23), splice d=0
5	TGCAGTGAAACCCGCGTGGA	-	47695204	1.285	10	1.0349	exonic(23/23)
6	TTACCAGGTAGGAGATTTAC	+	47655247	0.682	8	1.0317	exonic(19/23), splice d=0
7	CTAGCTCGTCGTCTGAAATG	+	47622820	0.952	14	1.0023	exonic(12/23), splice d=0
8	ACTAGCTCGTCGTCTGAAAT	+	47622819	0.749	10	0.9992	exonic(13/23), splice d=0
9	GCGGTACACTGCACTGTCCG	+	47619081	1.209	12	0.8589	exonic(23/23), splice d=11
10	AGCTAGTGAAGCGAGCCCAA	-	47622803	1.101	11	0.8011	exonic(23/23), splice d=6

All ten pass the hard specificity filter. Three structural readings. (1) The tier clusters: ranks 1-3 and 7-10 sit inside a 3.8 kb window spanning the validated functional region, so a multiplex construct carrying two or three of these guides is buildable without scaffolding problems, which matters for the escape-probability argument of Appendix J16. (2) The ranking is not an efficacy ranking: rank 5 has the second-highest efficacy in the tier (1.285) but sits mid-table because it lacks splice proximity; rank 6 is efficacy-weak (0.682) and survives on constraint. This is the v3 objective behaving as designed. (3) Ranks 7 and 8 are overlapping protospacers offset by one base - effectively one locus counted twice, a deduplication caveat we state rather than hide; in a build order they are alternatives, not a pair. The two filter-failing candidates (v2 ranks 10 and 14) and the two repetitive guides are excluded from this tier and preserved in Appendix K.

4. Discussion

4.1 Relation to existing tools

Existing gRNA design pipelines stop where this study begins. CHOPCHOP, CRISPOR, and CRISPick produce efficacy-and-specificity rankings for a user-supplied sequence; none asks what happens to the drive after it is released. Our contribution is to extend the design loop through population genetics: a candidate is not merely scored but simulated, and the simulation outcome

feeds back into the scoring rule (v2). To our knowledge this is the first published pipeline that ranks drive target sites against an experimentally validated positive control on the same quantitative axes the wet-lab study itself identified as decisive.

4.1.1 What this pipeline does not replace

It is equally important to state what the pipeline is not. It is not a replacement for CHOPCHOP or CRISPOR as interactive design front-ends - those tools are better for a user who wants one guide for one gene tonight, and our related-work comparison (Section 1.6) is about the drive-design intersection, not about displacing general-purpose tools. It is not a replacement for experimental off-target assays: GUIDE-seq, CIRCLE-seq and CHANGE-seq measure cleavage directly, including effects (cell type, chromatin, delivery) that no enumerator sees; our Stage-2 plan (Section 4.6) treats them as mandatory. It predicts resistance-generation rates only under stated assumptions - Section 4.9 builds per-candidate e intervals from microhomology inventories and an end-joining probability grid, and Limitation 5 owns the remaining gap between that prediction and a measured value. And it is not an argument that any drive should be released: the governance considerations of Section 4.4 stand independent of any ranking. The pipeline narrows the space of candidates that deserve expensive experiments; it does not argue the experiments away.

4.2 Limitations

The limitations are honest ones. (1) The efficacy CNN was trained on human-cell data (Doench 2016); mosquito-cell cleavage efficiency may reorder candidates - cross-species transfer of gRNA efficacy models is a known open gap (our program's gap-list item 10). The rankings are therefore design priors, not measured activities. (2) The off-target screen is exact over the AgamP5 reference haplotype; it does not model population-level sequence diversity (Ag1000G variation), so individual wild mosquitoes may carry private near-cognate sites the reference lacks. The AgamP4 mapping of Section 3.1.2 shows this is not hypothetical: the two named leads carry one and three mismatches respectively against the older reference, one of them at the PAM-adjacent seed base. The on-target half of that risk is now measured directly (Sections 4.7-4.8: 4,693 wild genomes at the three candidate windows); extending the enumerator to a pan-genome index, for genome-wide private near-cognate sites beyond those windows, remains the open extension. (3) The drive model is panmictic and allele-frequency only; spatial structure, density-dependent mating, and population suppression thresholds need an explicit population model before any release-relevant claim could be made. (4) Functional-constraint scoring is annotation-derived and coarse: the exon model itself is in question at exactly the site of interest (section 3.1), and resolving it required re-annotation against VectorBase AgamP4, now done (Appendix I). (5) The resistance-generation rate e is treated as a free parameter; in reality it is a measurable, sequence-dependent quantity (end-joining outcome spectra at a given cut site), and predicting it per candidate is now done under stated assumptions (Section 4.9); measuring it directly is the Stage-3 assay of Section 4.6.

4.2.1 Limitation 1: cross-species efficacy transfer

The CNN learned sequence-efficacy regularities from human-cell knockdown data. Cas9 biochemistry is conserved, but chromatin state, repair milieu and expression context are not, and mosquito-cell cleavage may reorder our candidates. We therefore never call the rankings "efficacious guides"; they are priors with a quantified same-assay advantage over a linear baseline (Appendix E) and a stated protocol caveat (Appendix N). The resolution is empirical: the Stage-1 in vitro panel of Section 4.6 measures mosquito-relevant cleavage directly, and the CNN can be fine-tuned on the resulting data - a few hundred measured guides suffice to recalibrate a ranking model.

4.2.2 Limitation 2: reference-genome specificity

All mm_i counts are exact over one haploid reference. Wild populations segregate variation the reference does not show, and a private 1-mismatch site in a release population would be invisible to our audit. The first stage of that measurement is now done: Section 4.7 screens 4,693 wild genomes at the three candidate on-target windows (withdrawing dsx-v3-1 at a 40.1% compromised-allele fraction), and Section 4.8 enumerates the population's near-cognate alleles at those windows. What remains open is the genome-wide half - private near-cognate sites outside the target windows - which requires the pan-genome index. Specificity claims outside the three windows still carry the qualifier "against AgamP5".

4.2.3 Limitation 3: the population model

The drive model is panmictic, deterministic in its main grid, and allele-frequency-only. It cannot express spatial structure, assortative mating, density dependence, or suppression thresholds, and its absolute timelines are optimistic by construction (Section 2.7.1). We use it only for comparative statements - which axis binds, which parameter to spend effort on - where the direction of the answer is robust to these omissions; any release-relevant claim would require an explicit spatial population model and is outside the scope of this paper.

4.2.4 Limitation 4: annotation-derived constraint

The v2/v3 constraint bonus is computed from the current RefSeq exon models, and those models are in dispute at exactly the site that matters most (Appendix I): the Kyrou guide is intronic under the current model and junction-spanning under the AgamP4-era annotation used in the paper. Our ranking does not depend on resolving the dispute - the bonus rewards any annotated constraint - but the dossiers' constraint labels would change under a corrected annotation, and the VectorBase AgamP4 re-annotation of Appendix I is the resolution, now done: the Kyrou guide is junction-spanning with 9 coding bp under the AgamP4 genebuild.

4.2.5 Limitation 5: e is a parameter, not a measurement

Section 3.4 shows the resistance-generation rate e is the decisive quantity, yet we sweep it instead of measuring it. In reality e is a sequence-dependent, assayable property of a cut site - the fraction of its end-joining spectrum that preserves gene function - and it differs across our own candidates.

Section 4.9 takes the first step: a microhomology-inventory predictor assigns each candidate an e interval and propagates it through the drive recursion. It remains a prediction, not a measurement - the end-joining probability anchor spans a 2,000-fold grid because the field has no Anopheles repair-outcome spectrum - and the Stage-3 assay of Section 4.6 measures e directly. Training indel-spectrum models on repair-outcome datasets remains the refinement that would sharpen the interval into a number.

4.3 Resistance-engineering strategies in the literature

Our result - that e , not h , is the binding constraint - aligns with and quantifies a design principle the field has converged on from several directions. The most direct mitigation is target choice under constraint, the Kyrou strategy this study operationalizes: cut where no functional indel exists. A second strategy multiplexes gRNAs against several nearby sites, so resistance requires simultaneous functional repair at every cut; modeling and cage data show multiplexing reduces effective e roughly multiplicatively, at the cost of larger drive constructs and more off-target exposure. A third recodes the drive's own copy of the target sequence so the drive allele is itself cleavage-resistant without relying on repair outcomes. A complementary line of work designs countermeasures - anti-drive elements that halt or reverse an unwanted drive - which matters for governance but does not change the design problem studied here. All of these strategies move effective e ; none moves h . Our pipeline's contribution is to make the first strategy quantitative, rankable, and testable before any construct is built.

4.3.1 A comparison of resistance-engineering strategies

strategy	mechanism	effect on effective e	cost	evidence base
constrained target choice (this work)	cut where every indel is deleterious	reduces functional-resistance fraction toward zero	requires a constrained site to exist	Kyrou 2018 cage collapse; our Sections 3.4, 3.6
multiplexed gRNAs	force simultaneous resistance at k sites	$e \rightarrow \sim e^k$ per event (Appendix J16)	larger construct, k -fold off-target exposure, correlated repair (J23)	modeling (Champer 2017), cage multiplex data
recoded drive allele	drive copy is cleavage-resistant by design	removes drive self-cutting	does not reduce wild-type resistance	standard drive-construction practice
anti-drive countermeasures	halt or reverse an unwanted drive	none on e ; governance layer	separate construct and release	modeling literature

The strategies compose rather than compete: a multiplexed construct at a constrained cluster (our Section 3.13.4 tier) with a recoded drive copy stacks all three e -reduction mechanisms, and the tiers were shortlisted with exactly that composition in mind.

4.4 Governance context

Gene drives are a governance problem as much as a technical one, and a computational study is the right place to say so plainly. The same property that makes drives attractive - self-propagation -

makes release effectively irreversible at current technology, so the field consensus is staged development: laboratory containment, caged population trials, and only then any consideration of field release, with community engagement and national regulatory approval at each step. This study operates entirely in silico on public sequence data, produces no engineered organism, and makes no release recommendation. Its purpose is the opposite of advocacy: to show quantitatively that naive drive designs fail, that the failure mode is predictable from population genetics, and that the difference between failure and the one published success lies in target-site biology that can and should be screened computationally before any cage is built.

4.5 What genome-wide enumeration changes

Exact genome-wide enumeration converts specificity from a heuristic score into an auditable fact. Heuristic tools report a summary score whose threshold choices are opaque; an exact count at each mismatch level is reproducible by any group from the same accession, and the count itself - not a model of the count - is what enters the design filter. Two consequences follow. First, the published guide's reputation for specificity is now a measured quantity (zero sites closer than 3 mismatches), which retrospectively de-risks the field's choice and prospectively defines the bar. Second, the observation that low-complexity protospacers accumulate hundreds of 3-mismatch sites (Figure 7, ranks 14 and 16) shows that 3-mismatch burden is an effective low-complexity detector, and argues for reporting the full mismatch histogram rather than a single thresholded count in any drive design study.

4.6 A staged validation plan for the named candidates

The dossiers of Section 3.13 are falsifiable, and this section states the cheapest falsification path. Stage 1 (in vitro): synthesize dsx-v3-1, dsx-v3-2 and the Kyrou guide; measure cleavage kinetics on the on-target dsx amplicon and on amplicons of the 13 enumerated 3-mismatch loci (4 + 9) plus the Kyrou guide's six - a 21-locus panel that fully covers every near-cognate site at ≤ 3 mismatches for all three guides. Decision gate: a lead advances only if on-target cleavage matches or exceeds the control and no panel locus cleaves above assay background. Stage 2 (cellular): GUIDE-seq or CIRCLE-seq in *Anopheles* cells for the two leads, testing whether the exact enumeration missed any site the assay detects (it should not, at ≤ 3 mismatches, by Appendix J1; discovery of such a site would falsify the enumerator, not just the candidate). Stage 3 (organismal): embryo microinjection with Cas9 protein and synthetic guide, scoring cut rates and indel spectra at the target - this also measures the resistance-generation rate e directly, the parameter Section 3.4 shows is decisive. Stage 4 (drive construct): insertion of a minimal drive at the lead site, germline conversion measurement, then small-cage population trials against the Kyrou drive design as control. Each stage is cheap relative to the next and each has a pre-stated kill criterion; the pipeline exists so that candidates die in silica or in vitro, not in cages.

4.7 From reference genome to population: the Ag1000G extension

Every specificity statement above is made against reference haplotypes, and wild populations are not the reference. This section converts that limitation into a measurement. From the Ag1000G phase-3 resource we fetched the per-sample VCF records spanning the three candidate target windows (AgamP4 2R:48,711,450-48,714,700, covering the dsx-v3-1, dsx-v3-2 and Kyrou protospacer+PAM units) for all 4,693 unique samples in the phase-3 manifest, using indexed remote queries (pysam/htslib) against the Sanger mirror. Calls were accepted at genotype quality ≥ 20 and depth ≥ 5 ; low-quality calls shrink the denominator rather than guess. Genotypes are unphased, so per-sample compromise calls are lower bounds: a sample heterozygous at two compromising sites is counted once even if both mutant alleles lie on one chromosome. The scan completed all 4,693 samples - 4,681 clean rows plus 12 per-sample fetch errors, counted and retained as error rows, never silently dropped - with row-level checkpointing across 18 interruptions; the operational log, including the worker-count experiment and the stall taxonomy, is Appendix R. Window coordinates and guide alleles were independently re-derived from the archived AgamP4 FASTA with Biopython before scoring (results/window_verification_biopython.json).

A chromosome is counted compromised for a guide if it carries any allele that disrupts the PAM GG dinucleotide or substitutes any protospacer base; variation at the PAM N position is tolerated, since SpCas9 requires only the GG dinucleotide (the cross-assembly audit of Section 4.10 uses the same 23-nt cleavage-relevant unit). The genotype accounting across the 4,693-sample cohort (denominator 9,386 chromosomes at fully called sites; results/ag1000g/allele_freq.json):

guide	intact	heterozygous	homozygous compromised	carrier fraction	compromised allele fraction
dsx-v3-1	2,158	1,306	1,229	0.540	0.401
dsx-v3-2	4,506	96	91	0.040	0.030
kyrou	4,506	82	105	0.040	0.031

dsx-v3-1's window is a polymorphism hotspot. The protospacer segregates common variants at multiple positions - 2R:48,711,511 T>A at allele frequency 0.466 (AC 4,236) and 2R:48,711,516 A>G at 0.286 (AC 2,644) head seven seed-region alleles - and the PAM N position carries a C>T variant at 0.527 (AC 4,772) that marks the alternative major haplotype. The per-sample accounting gives 40.1% of population chromosomes presenting a compromised target to a drive built on this guide, and we withdraw dsx-v3-1 as a candidate: a drive released at this site would face a resistant subpopulation before its first generation. The v3 composite scored a single reference haplotype; at this window no single reference represents the population, and the ranking could not see the haplotype it did not carry.

dsx-v3-2 tells the opposite story with the same machinery. Its one common variant (2R:48,712,765 A>G, AC 7,640, allele frequency 0.840) sits at seed position 0 - and the variant allele is the guide-matching one. The AgamP4 reference carries the minor allele at this position; AgamP5 and

the wild population carry the allele the guide was designed from. We call this the restoration allele: against AgamP4 the guide appears to carry a seed mismatch, and against the population it is exact. The cross-assembly replication of Section 4.10 confirms the direction independently. After accounting, 3.0% of chromosomes remain compromised, all at low-frequency protospacer and PAM-G sites.

The Kyrou window - the field's validated target - shows 3.1% compromised chromosomes and exactly three distinct single-variant alleles (Section 4.8), with no PAM variant of any kind in 9,386 chromosomes. For a junction-spanning essential target this is the expected purifying-selection signature, and it calibrates what 'clean' means quantitatively: the validated drive target is not variant-free, it is variant-poor at the cleavage-relevant positions.

Appendix J15's rule-of-three bound treated population near-cognates as a sampling problem; with the scan complete, the bound is replaced by a direct measurement at these windows. The population allele inventory is enumerated, not bounded.

4.8 The population's near-cognate allele inventory

Classifying every observed allele at the three windows by its cleavage consequence gives the first population-resolved near-cognate inventory for any drive target (results/ag1000g/near_cognate.json). AC is the chromosome count carrying each allele; a chromosome carrying several variants appears once per variant, so class totals are not additive to the 9,386-chromosome cohort.

guide	distinct single-variant alleles	PAM-G disrupted (alleles / AC)	PAM-N variant (tolerated)	seed mismatch	distal mismatch	wild-type	phase-certain multi-variant samples	ambiguous multi-site samples
dsx-v3-1	25	3 / 430	1 / 4,772	7 / 1,934	11 / 1,636	3 / 7,507	786	2,274
dsx-v3-2	15	2 / 11	-	4 / 191	8 / 203	1 / 7,640	85	62
kyrou	3	-	-	2 / 100	1 / 194	-	1	0

Three readings. First, the mechanism hierarchy is visible: dsx-v3-1's largest compromising classes are seed mismatch (AC 1,934 across 7 alleles) and distal mismatch (AC 1,636 across 11), with PAM-G disruption adding 430 - while the window's single most common variant (AC 4,772) is a tolerated PAM-N substitution that leaves the guide intact. Second, the Kyrou window carries zero PAM variants of any kind and only three protospacer-variant alleles (AC 100 seed, AC 194 distal) in 9,386 chromosomes: the purifying-selection signature that Section 1.3.1's design logic predicts for a junction-spanning essential target, now measured rather than assumed. Third, the 2,274 ambiguous dsx-v3-1 samples are an honesty boundary, not a defect: without phase information a sample heterozygous at two sites cannot be assigned one or two compromised chromosomes, and we report the lower-bound convention rather than impute phase. At the Kyrou window zero samples remain ambiguous in the full cohort - the one multi-site sample resolves phase-certain - a

change from the partial scan, stated so the zero is not mistaken for a gap. An MMseqs2 redundancy audit of the 43 allele 23-mers (87% identity, 0.9 coverage) collapses them to 27 sequence-similarity clusters, every multi-member cluster intra-guide: the three windows' inventories are sequence-independent, and dsx-v3-2's guide-matching restoration allele sits one cluster from its compromised neighbors (results/allele_clusters_mmseqs2.json).

4.9 Predicting the resistance-generation rate, and what the prediction says

Section 3.4 left e as a swept parameter; this section predicts it per candidate under stated assumptions and propagates the prediction through the drive recursion. The model is $e = p_{EJ} \times w$, where w is the fraction of microhomology-mediated deletions in the site's mutational target window ($L = 30$ bp around the cut) whose repaired product preserves gene function while destroying the protospacer match, and p_{EJ} is the probability that a failed homing event resolves by end-joining at all. The microhomology inventory is enumerated from sequence (results/resistance_rates.json): $w = 0.343$ (dsx-v3-1), 0.180 (dsx-v3-2), 0.211 (kyrou). The p_{EJ} grid spans $1e-4$ to 0.2 , anchored at the low end by the Kyrou cage experiments' zero observed functional resistance (PMC6871539) and at the high end by the Hammond 2016 drive's measured $0.4-8.6\%$ end-joining resistance rates (PMC4913862). Because standing variation is now measured, the recursion is initialized at each candidate's measured compromised-allele fraction r_0 from Section 4.7 rather than at $r_0 = 0$.

guide	w	r0 (measured)	predicted e interval	generations to 50% resistance (e low / e high)	generations to 95% drive
dsx-v3-1	0.343	0.401	3.4e-05 - 6.9e-02	11 / 11	never reached
dsx-v3-2	0.180	0.030	1.8e-05 - 3.6e-02	12 / 12	never reached
kyrou	0.211	0.031	2.1e-05 - 4.2e-02	12 / 12	never reached

The consequence is the study's sharpest result, and we name it the sieve law. Under full homozygous sterility ($c_{hom} = 1$), homing preferentially converts wild-type chromosomes while compromised chromosomes evade cleavage, and the sterile drive-homozygote class purges drive alleles every generation; the resistant class is therefore sieved upward through the breeding pool. At every computed point - three measured r_0 values crossed with both ends of a 2,000-fold p_{EJ} grid - R reaches 50% allele frequency within 11-12 generations, and no run sustains 95% drive frequency. The end-joining rate, decisive when $r_0 = 0$ (Section 3.4), becomes second-order once standing variation is measured: the population gate r_0 , not predicted efficacy and not e , is the first filter in drive target design. Within the model the law generalizes to any $c_{hom} = 1$ drive with $r_0 > 0$; outside the model it is stated for the three measured candidates only.

The claim is falsifiable twice over. The r_0 inputs are remeasurable from the public per-sample accessions (Appendix Q); the recursion is re-runnable, and results/resistance_rates.json records every input (w , L , r_0 , p_{EJ} grid) and every output. And the law would be wrong if a $c_{hom} = 1$

drive with $r_0 \sim 0.03$ kept functional resistance rare past twice the predicted horizon (24 generations) in a caged trial.

4.10 Cross-assembly replication on AgamP5

Every window coordinate, specificity audit, and allele-frequency claim above is made on AgamP4. A reference assembly is one haplotype, and an assembly error or a minor reference allele at the target would silently propagate through every downstream number. We therefore re-checked the three guide targets on the independent AgamP5 reference (RefSeq GCF_943734735.2; chromosome 2 NC_064601.1, chromosome 3 OX030908.1, X OX030909.1, mitochondrion OX030910.2), searching each molecule on both strands for the 23-nt protospacer+PAM unit by exact substring match and a pigeonhole near-match filter (four blocks; any ≤ 3 -mismatch hit preserves at least one intact block), with no aligner involved (scripts/cross_assembly_verify.py; results/cross_assembly_verify.json).

Three findings. First, the Kyrou target replicates exactly: the guide unit is present once in the AgamP5 genome, at chromosome 2:47,622,174, with zero secondary sites at ≤ 3 mismatches genome-wide over the full 23-mer. The zero-resistance and zero-secondary-site claims for this guide therefore hold on a second, independently assembled haplotype. Second, dsx-v3-2's guide unit is likewise exact and unique (chromosome 2:47,620,304), but the AgamP4 reference allele at that window is absent from AgamP5 even at ≤ 3 mismatches: AgamP4 carries a minor allele at this position, and AgamP5 carries the guide-matching allele that dominates the Ag1000G population samples (the restoration allele of Section 4.7). The assembly-mismatch caveat of Section 3.1.2 is thus confirmed from the other direction. Third, dsx-v3-1's guide unit is present at chromosome 2:47,619,040 with a single substitution at the PAM N position (tolerated; SpCas9 requires only the GG dinucleotide), while the AgamP4 reference allele - which differs from the guide at four protospacer positions - is again absent from AgamP5. AgamP4's reference haplotype is the outlier at both polymorphic windows, not the population and not the second assembly.

A methods note on consistency. The earlier genome-wide audit of the Kyrou guide (Appendix L) counted six 3-mismatch sites against the 20-nt protospacer alone; the present 23-mer audit counts zero secondary sites. These are complementary, not contradictory: we verified that the chromosome-2 exemplar of those six (NC_064601.1:41,285,162) carries an AAC sequence at the PAM position and scores six mismatches over the full 23-mer - protospacer-level near matches exist, but none brings a compatible PAM within the same mismatch budget. The protospacer+PAM unit is the cleavage-relevant object, and on that bar all three guides are genome-wide unique on AgamP5.

Two orthogonal toolchains confirm the coordinate and uniqueness claims. minimap2 (asm5, an aligner with gap placement, unlike our exact-substring search) maps the 90 kb AgamP4 region onto AgamP5 2RL as ten collinear blocks with no inversions and places all three guide windows within 7-12 bp of the aligner-free coordinates; the uniform small deltas are gap-placement effects

of the chained alignment, largest at the Kyrou window, consistent with the indel-rich RB-exon region of Appendix I (results/cross_assembly_minimap2.json). And NCBI BLAST+ (blastn-short against a local database of the four archived AgamP5 molecules) finds exactly one ungapped full-length 23-mer hit per guide genome-wide - its own site at the verified coordinate - and zero full-length hits at any non-zero mismatch count, independently reproducing the 23-mer uniqueness result (results/offtarget_blast.json). And Bowtie (-v 3, full-length ungapped ≤ 3 -mismatch search - the mode whose semantics match the audit's bar exactly) reports exactly three alignments genome-wide: the three on-target sites at the verified coordinates, with zero secondary sites (results/offtarget_bowtie.json). The aligner-free enumerator remains the base-precise claim; the three aligner-based tools bound the risk that its exactness assumption hid a systematic miss.

A fourth orthogonal check is structural rather than alignment-based. ViennaRNA 2.7.2 (MFE folding of each 20-nt spacer as RNA, the species loaded into Cas9) against a null of 500 random 2R 20-mers (mean MFE -1.104 kcal/mol, sd 1.703) flags dsx-v3-3 as a structural outlier: its spacer folds at -4.70 kcal/mol, $z = -2.11$ against the genomic null, while dsx-v3-1 (-1.20, $z = -0.06$), dsx-v3-2 (0.00, $z = +0.65$) and the Kyrou control (-1.80, $z = -0.41$) sit within the null range (results/guide_structure_vienna.json). Stable spacer self-structure is a documented efficacy confound, so this is a QC caveat on v3-3, the third-ranked lead; the two leads that survived the population screen are unaffected.

A fifth check is a fully independent curated pipeline. The 4,801-bp dsx window (AgamP5 NC_064601.1:47,618,500-47,623,300, fasta input) was submitted to CHOPCHOP v3 (Cas9 knock-out preset, NGG, 20-mers), which ranked 585 window guides against its own AgamP4 index (results/offtarget_chopchop.json). The three leads and the Kyrou control are all recovered with specificity profiles consistent with ours: dsx-v3-1 has zero MM0-MM2 matches in AgamP4 (2 MM3), dsx-v3-2 a single MM1 near cognate, and the Kyrou guide exactly one MM0 - its own site, conserved across the two assemblies - with nothing nearer than MM3. CHOPCHOP's efficiency ordering (Kyrou 64.48 > dsx-v3-1 63.74 > dsx-v3-2 54.90 > dsx-v3-3 54.08) also matches ours. One new caveat: dsx-v3-3 returns MM0=2 in AgamP4 - an exact-match duplicate exists in that assembly although the spacer is unique in AgamP5 under both Bowtie and BLAST+ - so the third lead carries an assembly-dependent copy-number flag on top of its structural one; both flagged issues attach to the lead that already failed to survive the population screen, leaving the v3-1/v3-2 pair untouched.

A sixth check repeats the curated-pipeline exercise on a second independent service and adds a cross-species dimension. CRISPOR (Haeussler et al. 2016) caps pasted-sequence input at 2,300 bp, so the dsx region was submitted as two windows (W1 AgamP5 NC_064601.1:47,618,700-47,620,999 carrying dsx-v3-1 and dsx-v3-2; W2 47,622,000-47,623,000 carrying the Kyrou control and dsx-v3-3; all four spacers verified present in the submitted sequence) against two curated genomes: AgamP4 (EnsemblMetazoa 76), the assembly CHOPCHOP used, and Anopheles coluzzii AcolN3 (GCF_943734685.1)

(results/offtarget_crispor.json). On AgamP4 the two engines agree for three of the four guides: dsx-v3-1 is again clean through 2 mismatches (2 MM3), the Kyrou control shows nothing nearer than MM4 once counting conventions are aligned (CHOPCHOP's MM0=1 is the conserved own site, which CRISPOR excludes by construction), and dsx-v3-2's single MM1 is recovered - and is resolved by its coordinates, 2R:48,712,762-48,712,784 in the dsx exon: it is the guide's own target locus, where the AgamP4 reference carries a PAM-distal position-20 C>T allele (mit 41.70, CFD 0.300), a reference-assembly allele difference already recorded in our reannotation (results/agamp4_reannotation.json), not a dispersed off-target. The engines disagree on dsx-v3-3: CHOPCHOP's index returned an exact-match duplicate (MM0=2) while CRISPOR's independent AgamP4 index returns MM0=0, so the copy-number flag is downgraded from confirmed to engine-specific and unresolved; it remains a caveat on the lead the population screen already withdrew. The coluzzii run is the first cross-species check in this study: all four on-target sites are recovered exactly at the coluzzii dsx ortholog, with dsx-v3-2 (MIT specificity 100, CFD 100) and the Kyrou guide clean through 3 mismatches and dsx-v3-1 and dsx-v3-3 clean through 2 - the target sites are conserved across the two sympatric, hybridizing members of the gambiae complex, and the coluzzii reference matches the guide allele at the position where AgamP4 disagrees.

A seventh check changes method family entirely: CRISPROff/CRISPRspec (v1.1.2, RTH-tools; Alkan et al. 2018, Genome Biol 19:177) scores specificity from RNA/DNA duplex thermodynamics rather than alignment counts or ML models. The pipeline was validated against its own published example outputs (repo test.sh reproduces test_data.example.out diff-clean) before use, with its RNAfold requirement met by the ViennaRNA 2.7.2 bindings already used in this study, and run on the CRISPOR AgamP4 near-cognate sets of the four leads (results/offtarget_crisproff.json). The resulting CRISPRspec ranking is Kyrou control (14.57) > dsx-v3-1 (11.95) > dsx-v3-3 (11.66) > dsx-v3-2 (3.72). The energy model independently singles out exactly the site the cross-engine check resolved: dsx-v3-2's single one-mismatch near-cognate - its own dsx locus carrying the AgamP4 position-20 C>T allele - carries a per-site CRISPROff score of 18.09, 77% of the on-target binding energy (23.37), quantifying how strongly that variant allele would bind where it occurs and corroborating the population-screen withdrawal of dsx-v3-2 on a third, physics-based axis.

An eighth check scales the audit from the four leads to every possible guide in the dsx windows: FlashFry (v2.0; McKenna and Shendure 2018, BMC Biology, doi:10.1186/s12915-018-0545-0) builds a whole-genome AgamP5 off-target database (all four molecules, 209 MB, built in 192 s) and enumerates off-targets at up to four mismatches for all 430 Cas9-NGG guides discoverable in the two windows, scoring each with CFD, Hsu 2013 and Doench 2014 metrics (results/offtarget_flashfry.json, results/flashfry_scored_guides.tsv). Across the 421 scored guides, genome-wide hit totals grow geometrically with mismatch count (15 at one mismatch, 171 at two, 1,491 at three, 11,276 at four) and no guide in either window is free of off-targets at four mismatches. For the leads, FlashFry resolves the remaining open flag: dsx-v3-2 shows zero

one-to-three-mismatch off-targets in AgamP5 (mismatch histogram 1,0,0,0,3) - the one-mismatch site CRISPOR reported and CRISPROff weighted at 77% of on-target energy is specific to the AgamP4 reference allele and absent from AgamP5, so the flag tracks the assembly rather than the guide. dsx-v3-2 is in fact the most sequence-specific lead on this axis (CFD specificity 0.782), while dsx-v3-3 is the least (0.567; ten off-targets at four or fewer mismatches, two at three), a third independent caution on that guide after the ViennaRNA structural outlier. No lead carries a multi-hit exact-match duplicate in AgamP5, agreeing with CRISPOR against CHOPCHOP's engine-specific duplicate flag.

A ninth check replays the library-scale audit with a second, independent FM-index aligner: bowtie2 2.5.4 (Langmead and Salzberg 2012) aligns all 421 unique 20-mer protospacers against the whole AgamP5 genome in exhaustive all-hits mode with a three-mismatch, ungapped scoring budget (results/offtarget_bowtie2.json). The exercise yields two honest findings beyond confirmation. First, a sensitivity negative result: bowtie2's default --sensitive preset (22 bp seeds) silently under-reports the off-target space for 20-mer queries, returning 660 alignments and zero two-to-three-mismatch sites, because a 22 bp seed cannot tolerate two or three mismatches inside a 20-mer read; the of-record run uses short 14 bp seeds with mismatch-tolerant seeding and hundred-fold extension effort (24,278 alignments), a concrete demonstration that default short-read settings are unsafe for guide off-target enumeration. Second, a PAM-agnostic census: the library's mismatch histogram grows 427 / 227 / 2,744 / 20,880 at zero to three mismatches, roughly fourteen times FlashFry's PAM-adjacent counts at every mismatch level - a direct measurement of how strongly the NGG requirement filters near-cognate sequence in this genome. All four leads remain free of any one-or-two-mismatch near-cognate even PAM-agnostically. The census also refines the duplicate story: three of 421 guides (0.7%) carry perfect 20-mer copies at other genomic loci (six copies in total), but none of the six has an adjacent NGG PAM, so every copy is Cas9-inert - FlashFry's zero-MM0-duplicate result for PAM-adjacent sites is confirmed, and CHOPCHOP's duplicate flag stays engine-specific. (This census is later shown to under-report the highest-mismatch tiers; the eleventh check below corrects the census of record to 32,305.)

A tenth check asks the functional question behind the census: do any of the 24,278 PAM-agnostic near-cognate sites overlap protein-coding sequence? bedtools 2.31.1 intersect (Quinlan and Hall 2010) of the full alignment set against the AgamP5 RefSeq annotation (GCF_943734735.2, release RS_2023_12; 206,438 CDS features) shows that roughly one in eight one-to-three-mismatch near-cognates genome-wide overlaps a CDS (29/227 at one mismatch, 390/2,744 at two, 2,624/20,880 at three), and 239 of 421 library guides carry at least one such site - a library-scale measure of the background the NGG filter and mismatch intolerance protect against (results/offtarget_context_bedtools.json). For the four leads the audit is clean after inspection: dsx-v3-1 and dsx-v3-2 have no CDS-overlapping near-cognate at any tier; dsx-v3-3 has exactly one three-mismatch site inside LOC1270942 (VPS13B, chr2:38,008,162-38,008,181) and the Kyrou control one inside LOC1274366 (chr2:41,285,163-41,285,182), and neither site carries an

adjacent NGG (or reverse-orientation CCN) in AgamP5 - flanks GCG|GTG and GTT|TAC respectively - so both are Cas9-inert and the leads' safety picture is unchanged, now with gene-level annotation behind it. (The eleventh check below found the underlying census under-counted and re-ran this context analysis on the corrected 32,305-alignment set; every lead-level conclusion here is unchanged.)

An eleventh check reruns the enumeration with BWA 0.7.19-r1273 (Li and Durbin 2009) - the engine inside CRISPOR, run here first-party - in its exhaustive non-iterative mode with gaps disabled, a pure-Hamming census of the same 421 guides against AgamP5 (results/offtarget_bwa.json). The exercise produces an uncomfortable, valuable correction. BWA returns 32,305 ungapped zero-to-three-mismatch alignments (427 / 227 / 2,886 / 28,765), a strict superset of the bowtie2 census: zero of the 24,278 bowtie2 alignments are missing, and 8,027 additional alignments - all at two or three mismatches - appear only in BWA. Fifty of those bwa-only sites, sampled at fixed seed, were re-verified by direct reference extraction and Hamming recount, and all fifty are true three-mismatch-or-better sites. The bowtie2 census of record therefore under-reports the near-cognate space by 25% (by 38% at three mismatches): even after seed tuning, bowtie2's alignment-score threshold silently trims high-mismatch ungapped hits. The corrected census of 32,305 is adopted; the bowtie2 table is retained above as a documented under-count, and the lesson of the ninth check is hardened - aligner recall for guide enumeration must be verified against a second engine, not against the same engine's settings sweep. The correction propagates to the functional-context audit: re-run on the corrected alignment set, one-to-three-mismatch CDS overlap becomes 29/227, 412/2,886 and 3,452/28,765 (12.0-14.3% by tier), and 308 of 421 library guides now carry at least one CDS-overlapping near-cognate. The superset adds no new one-mismatch site anywhere, no new CDS-overlapping site for any lead, and the two known lead CDS overlaps (dsx-v3-3 in VPS13B, Kyrrou in LOC1274366) stand as resolved - both PAM-less, both Cas9-inert. Every lead-level conclusion of the ninth and tenth checks survives the correction intact.

5. Conclusion

A CNN + population-genetics pipeline recovers the field's validated dsx drive gRNA, nominates 93 higher-scoring candidates (v1 top: ACGGACGGATACAGACTGGA-CGG), and shows quantitatively that resistance generation - controllable through target-site choice - is the decisive design variable for sterility-inducing mosquito gene drives. The constraint-aware v2 ranking (top: GCGGTACACTGCACTGTCCG-CGG) independently concentrates on the functional region the field validated. A specificity-fused v3 ranking on exact genome-wide enumeration nominates two named candidates (dsx-v3-1, dsx-v3-2) that outscore the published guide on the fused composite, while the first exact genome-wide audit of that guide quantifies the specificity bar the field implicitly set. A 4,693-genome population screen then withdraws dsx-v3-1 (40.1% of wild chromosomes compromise its target) and confirms dsx-v3-2 (3.0%), and a microhomology-based e

predictor coupled to the measured standing variation yields the sieve law: resistant alleles reach 50% frequency within 11-12 generations regardless of end-joining rate, making the population gate the first design filter. The next iterations are pan-genome near-cognate enumeration beyond the target windows and direct measurement of end-joining outcome spectra at the lead site.

5.1 Summary of contributions

1. A complete, reproducible design pipeline for dsx drive protospacers: scanner, CNN efficacy model, exact off-target enumeration at region and genome scale, constraint-aware and specificity-fused rankings, and a three-allele drive simulator, all pinned by a 77-test hermetic suite.
2. The first exact genome-wide specificity audit of the published Kyrou 2018 guide: 0 sites at ≤ 2 mismatches, exactly 6 at 3 mismatches, coordinates enumerated (Appendix L).
3. A named, falsifiable shortlist: dsx-v3-1 and dsx-v3-2, with complete off-target dossiers (13 enumerated 3-mismatch loci), scored above the published control on the fused v3 composite (Section 3.13), plus a staged validation plan with pre-stated kill criteria (Section 4.6). The population screen of Section 4.7 then withdraws dsx-v3-1 (40.1% standing compromised alleles), leaving dsx-v3-2 as the single lead - the shortlist's falsifiability was exercised, not just stated.
4. A quantitative design law from the parameter grid: homing rate moves peak drive frequency, resistance-generation rate moves the outcome; target constraint is the cheapest available lever on the latter (Sections 3.4, 3.4.1).
5. The annotation discrepancy of the original Appendix I resolved: the Kyrou guide is junction-spanning with 9 coding bp under the AgamP4/VectorBase genebuild, and the leads are quantifiably assembly-sensitive (Section 3.1.2).
6. Honest negatives, preserved: the drive fails in 15 of 16 simulated cells; the enumerator loses to published Cas-OFFinder throughput by 10^2 - 10^4 (Appendix M); the CNN's advantage is demonstrated only against a matched ridge baseline, not against Rule Set 2 (Appendix N); two candidates fail the specificity filter and two more are repetitive (Appendix K); the RefSeq annotation contradicts the published target description at the exact site of interest (Appendix I).
7. The first population-scale measurement of drive-target standing variation: compromised-allele fractions at three candidate windows across all 4,693 genomes of the Ag1000G phase-3 manifest (4,681 clean plus 12 accounted error rows), with the near-cognate allele inventory classified by cleavage consequence (Sections 4.7-4.8).
8. A named, quantified, falsifiable discovery - the sieve law: with full homozygous sterility and measured standing variation $r_0 > 0$, resistant alleles reach 50% frequency within 11-12 generations at both ends of a 2,000-fold end-joining grid, and 95% drive frequency is never reached; the population gate, not efficacy or e , is the first filter (Section 4.9).

5.2 What would make this paper wrong

The claims are falsifiable, and we state their falsifiers. The enumerator is wrong if any ≤ 3 -mismatch site exists that our pigeonhole filter missed - detectable by any independent exact enumerator over the same accessions (the completeness proof of Appendix J1 makes a miss a proof bug, not a tuning issue). The candidate claims are wrong if the Stage-1 panel of Section 4.6 shows weak on-target cleavage or strong cleavage at any of the 13 enumerated loci. The design law is

wrong if a drive with high e and moderate h outlasts a low- e high- h drive in a controlled cage comparison. The annotation finding is wrong if the AgamP4 re-annotation reproduces the junction-spanning model under the current record. Each falsifier is cheap to execute relative to the claim it tests, which is the point of writing the paper this way.

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Appendix A. Data provenance

All genomic inputs are archived in the repository under data/: dsx_locus_region.fasta (NC_064601.1:47600000-47710000, fetched 24 Sep 2026 via NCBI efetch), dsx_gene_table.txt (RefSeq gene table, nine transcript variants), dsx_gene_esummary.json (NCBI Gene 1270904 record documenting aliases AGAP004050/DSX), dsx_transcripts.fasta (transcript sequences). The Kyrou gRNA sequence was extracted from the PMC6871539 full text (Europe PMC API). Efficacy training data (Doench 2016 FC+RES, 5,310 guides; Doench V1, 2,144 guides) come from the Microsoft Research azimuth mirror and are documented in the item-6 groundwork archive. No input is synthetic.

A.1 Artifact inventory

Every result file is regenerated by a named script from archived inputs:

artifact	produced by	contents
results/analysis_summary.json	mosqdesign.run_analysis	v1 scan, efficacy distribution, control recovery
results/ranked_designs.csv	mosqdesign.run_analysis	v1 top-150 ranked candidates
results/analysis_v2_summary.json	mosqdesign.run_analysis_v2	v2 constraint analysis summary
results/ranked_designs_v2.csv	mosqdesign.run_analysis_v2	v2 top-300 candidates with constraint labels
results/wright_fisher_traj.npy	mosqdesign.drive_sim	stochastic trajectories (5 seeds)
results/genome_offtargets_chr2_top30.{csv,json}	scripts/run_genome_scan.py	2RL-scale enumeration (superseded)

results/genome_offtargets_genomewide.{csv,json}	scripts/run_genome_scan.py	full AgamP5 enumeration, per-hit coordinates
results/kyrou_genomewide_audit.json	scripts/run_genome_scan.py	published-guide audit (Appendix L)
results/ranked_designs_v3.csv	mosqdesign.v3_rank	v3 fused ranking of 30 candidates
results/v3_summary.json	mosqdesign.v3_rank	v3 summary and control comparison
data/agamp4/AGAP004050_AgamP4.gff3	scripts/fetch_agamp4.sh	archived VectorBase genebuild excerpt
data/agamp4/AgamP4_2R_dsx_region.fasta	scripts/fetch_agamp4.sh	archived AgamP4 2R dsx region
results/agamp4_reannotation.json	scripts/agamp4_reannotation.py	junction-spanning resolution + lead mapping
figures/fig1-fig10	mosqdesign.make_figures + paper scripts	all figures
paper/paper_draft.pdf	scripts/build_paper.py	this document

Appendix B. Reproducibility

From a clean checkout, the complete pipeline is:

```

pip install -e . # install the mosqdesign CLI
scripts/fetch_data.sh # archive locus FASTA, gene table,
esummary (network)
scripts/fetch_genome.sh # archive the four AgamP5 molecules
(network)
mosqdesign run-v1 # v1: scan, score, rank, simulate
mosqdesign run-v2 # v2: constraint-aware ranking
python3 scripts/run_genome_scan.py # genome-wide enumeration
(top-30 + Kyrou audit)
mosqdesign rank-v3 # v3 specificity-fused ranking
scripts/fetch_agamp4.sh # archive AgamP4 evidence (network)
python3 scripts/agamp4_reannotation.py # AgamP4 re-annotation of the
Kyrou site
python3 -m pytest # 77-test hermetic suite
python3 scripts/build_paper.py # render this PDF

```

scripts/fetch_data.sh downloads and archives all external inputs (the only step requiring network); the analysis scripts regenerate results/analysis_summary.json, results/analysis_v2_summary.json, results/ranked_designs.csv, results/ranked_designs_v2.csv, and the figure set (figures/fig1-fig10); python3 -m pytest runs the 77-test hermetic suite with no network access; scripts/build_paper.py renders this PDF. The package installs a `mosqdesign` console script with four subcommands: run-v1 (locus scan, CNN scoring, v1 ranking, drive grid), run-v2 (constraint-aware ranking), rank-v3 (specificity-fused ranking; regenerates results/ranked_designs_v3.csv and results/v3_summary.json byte-identically, a property pinned by the test suite), and offtargets (exact $\leq N$ -mismatch enumeration of any protospacer against any FASTA). As a worked control, `mosqdesign offtargets --proto GTTTAACACAGGTCAAGCGG --fasta data/dsx_locus_region.fasta --max-mismatches 2` reproduces the Section 3.1 result on the archived locus: exactly one exact hit (the guide's own site) and zero hits at 1-2 mismatches. All stochastic components are seeded. The environment requires only Python 3, NumPy, PyTorch, matplotlib, and reportlab, and runs within a 1 GB RAM / 2 CPU sandbox.

Appendix C. Verification suite

module	test	what it pins
test_grna_scan	test_finds_plus_strand_site	plus-strand NGG detection
test_grna_scan	test_finds_minus_strand_site	minus-strand NGG detection
test_grna_scan	test_context_padding_at_edges	N-padding of context windows at region edges
test_grna_scan	test_no_ambiguous_bases_in_sites	rejection of ambiguous-base windows
test_grna_scan	test_homopolymer_run	homopolymer-run measurement
test_grna_scan	test_count_offtargets_excludes_self	self-match exclusion in off-target counts
test_grna_scan	test_build_kmer_index_and_fast_offtargets	5-mer index construction
test_grna_scan	test_fast_matches_naive_on_random_panel	pigeonhole == exhaustive scan on random panels
test_efficacy_scorer	test_scorer_outputs_finite_scores	CNN output finiteness/shape
test_efficacy_scorer	test_one_hot_basic	one-hot encoding correctness
test_rank	test_composite_prefers_high_efficacy_no_offtarget	composite ordering
test_rank	test_gc_penalty	GC-band penalty
test_rank	test_rank_orders_by_score	ranking consistency
test_annotation	test_gene_table_parsing_all_variant_tables	all nine transcript exon models parsed
test_annotation	test_kyrou_site_annotation_discrepancy_documented	the Kyrou intronic-vs-junction discrepancy stays visible
test_annotation	test_constraint_bonus_logic	v2 exonic/splice bonus rules
test_drive_sim	test_allele_freqs_sum_to_one	conservation of allele frequencies
test_drive_sim	test_drive_invades_from_rare	drive invasion from 1%
test_drive_sim	test_no_homing_no_spread	no spread at $h = 0$
test_drive_sim	test_high_resistance_rate_blocks_drive	resistance blocks the drive at high e
test_drive_sim	test_full_sterile_female_drive_suppresses_like_kyrou	qualitative Kyrou cage-collapse regime
test_drive_sim	test_wright_fisher_shape_and_seed	seeded stochastic reproducibility

| test_genome_scan | test_iter_fasta_blocks_covers_every_window | block overlap covers every 20-mer start |

| test_genome_scan | test_iter_fasta_blocks_flushes_final_partial_block | final partial block is not dropped |

| test_genome_scan | test_scan_matches_naive_counter_with_boundary_hit | vectorized scan matches naive counter at block boundary |

| test_genome_scan | test_no_double_count_across_overlap | overlap region yields no double counts |

| test_genome_scan | test_multiple_guides_independent_counts | guides do not contaminate each other's counts |

| test_genome_scan | test_numpy_path_matches_pure_python_path | numpy path identical to pure-Python reference |

| test_genome_scan | test_n_bases_never_match | ambiguous bases never verify as hits |

| test_genome_scan | test_hit_at_sequence_edge | hits at block edges are found |

| test_v3_rank | test_filter_rejects_any_exact_or_single_mismatch_hit | hard filter on mm0 + mm1 |

| test_v3_rank | test_penalty_scales_with_mismatch_closeness | 0.50 per mm2, 0.05 per mm3 |

| test_v3_rank | test_score_v3_combines_terms | v3 = efficacy + bonus - penalties |

| test_agamp4 | test_kyrou_site_junction_spanning_under_agamp4 | 9-bp coding overlap at the AgamP4 RB exon boundary |

| test_agamp4 | test_female_exon_135bp_present | 135-bp female exon at 2R:48,715,161-48,715,295 |

test_agamp4	test_leads_are_assembly_sensitive	dsx-v3-2 seed-base mismatch, dsx-v3-1 3 mismatches vs AgamP4
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The suite has since grown to 77 tests. The eight modules tabulated above (36 tests) pin the original pipeline; eight further modules (41 tests) pin the extensions of Sections 4.7-4.10, the command-line interface, and their supporting checks:

module	what it pins
test_ag1000g_af (8 tests)	population-scan aggregation: role assignment, multiallelic splitting, quality-filter denominator shrinkage, PAM-G versus PAM-N consequence, unphased lower-bound calling, summary accounting, window bases
test_cross_assembly (8 tests)	reverse complement, 23-mer block coverage, Hamming distance, overlapping exact and near search, orientation conventions
test_guide_fold_primers (3 tests)	deterministic folding, amplicon spans the window, report consistency
test_near_cognate (8 tests)	allele classification (seed/distal/PAM), single-variant enumeration, phase-certain versus ambiguous multi-site samples, summary counts
test_resistance_rate (6 tests)	cut-site geometry, mutational target size, microhomology detection, resistance classification, region-FASTA end-to-end run, window bounds

test_uniprot_isoforms (2 tests)	archived sex-specific isoform records, distinct positive lengths
test_cli (3 tests)	CLI dispatch, synthetic-FASTA off-target enumeration, archived-locus Kyrou control reproduction, byte-identical v3 artifact regeneration
test_window_verification (3 tests)	Biopython re-derivation of all scan windows from the archived AgamP4 FASTA, Kyrou exact match, PAM strand convention

Appendix D. Constraint-aware (v2) top-30 candidates

rank	protospacer	PAM	strand	genomic pos	efficacy	constraint bonus	off-targets	v2 score
1	GCGGTACACTGCACTGTCCG	CGG	+	47619081	1.2089	0.75	0	1.9589
2	AGCTAGTGAAGCGAGCCCAA	TGG	-	47622803	1.1011	0.75	0	1.8511
3	GCACACCAGCGGATCGACGA	AGG	-	47622696	0.9941	0.75	0	1.7441
4	TACTTACCTCCCATATGCTG	CGG	+	47692033	0.9579	0.75	0	1.7079
5	CTAGCTCGTCGTCTGAAATG	GGG	+	47622820	0.9523	0.75	0	1.7023
6	GAAGCGAGCCCAATGGCTGT	TGG	-	47622796	0.9367	0.75	0	1.6867
7	GAACCTGTAAATCTCCTACC	TGG	-	47655250	0.9013	0.75	0	1.6513
8	CATTAAGACCTACGAAGCGC	TGG	-	47620304	0.9004	0.75	0	1.6504
9	TGCGGCTCGGGAACGTGGTG	CGG	+	47692051	0.8804	0.75	0	1.6304
10	CAGTGTAACCGCTGTACAAAA	AGG	-	47619069	0.9725	0.75	0	1.6225
11	ATGTACAAAAAGCACACCAG	CGG	-	47622707	1.0716	0.75	0	1.6216
12	ACGGACGGATACAGACTGGA	CGG	+	47619412	1.3447	0.25	0	1.5947
13	TCGAGGAAGCGATGGTGCTA	GGG	+	47693451	0.8406	0.75	0	1.5906
14	GCTTACCTTCGTGATCCGC	TGG	+	47622691	0.8302	0.75	0	1.5802
15	TGCAGTGAAACCCGCGTGGA	CGG	-	47695204	1.2849	0.25	0	1.5349
16	GCGGATCGACGAAGGTAAGC	TGG	-	47622688	0.7563	0.75	0	1.5063
17	CCCGAGCCGCAGCATATGGG	AGG	-	47692039	0.7501	0.75	0	1.5001
18	ACTAGCTCGTCGTCTGAAAT	GGG	+	47622819	0.7492	0.75	0	1.4992
19	ACACTTACCTTTTTGTACAG	CGG	+	47619062	0.9353	0.75	0	1.4853
20	GTGCGAAGCGCACGAAACAC	TGG	+	47700905	0.7108	0.75	0	1.4608
21	CCGTTGCAGTGAAACCCGCG	TGG	-	47695208	1.2027	0.25	0	1.4527
22	ACCTCTCCGAACGTTTGCAT	TGG	-	47700884	0.7022	0.75	0	1.4522
23	CACTAGCTCGTCGTCTGAAA	TGG	+	47622818	0.6981	0.75	0	1.4481
24	TATGCTGCGGCTCGGGAACG	TGG	+	47692046	0.6975	0.75	0	1.4475
25	TTACCAGGTAGGAGATTAC	AGG	+	47655247	0.6817	0.75	0	1.4317
26	GGTAGGAGATTACAGGTTC	TGG	+	47655253	0.6784	0.75	0	1.4284
27	GTCGAGGAAGCGATGGTGCT	AGG	+	47693450	0.6765	0.75	0	1.4265

28	TGGGCAGTATGCGTTAGGGT	AGG	-	47619040	0.9248	0.50	0	1.4248
29	ACGGTGGCGCGCGAGCGAGA	CGG	-	47696369	0.6718	0.75	0	1.4218
30	TGTGTCGTTTACCTTTTGG	GGG	+	47620325	0.8650	0.75	0	1.4150

Exonic-only candidates (bonus 0.25) are visible at ranks 12, 15, 21, and 31 in the full table (results/ranked_designs_v2.csv); the v1 leader is among them.

Appendix E. Efficacy model evaluation detail

evaluation	guides	CNN Spearman	ridge Spearman	CNN advantage
held-out FC+RES test split (20%, seed 0)	1,062	0.556	0.463	+0.093
cross-dataset Doench V1 (no retraining)	2,144	0.609	0.444	+0.166

Training set: 4,248 guides (80% of FC+RES, seed 0), 15 epochs, Adam, MSE loss, no early stopping or hyperparameter selection on the test split. The cross-dataset row is the more conservative estimate of real-world ranking quality, because it evaluates transfer across experimental assays - a weaker proxy of the human-to-mosquito transfer actually required, which remains the study's main stated limitation.

Appendix F. Trajectory of the single successful grid cell

Deterministic trajectory for ($h = 0.99$, $e = 0.001$, $c_{\text{hom}} = 1$, $s_{\text{het}} = 0$), drive introduced at 1% - the only cell that crosses 95% drive frequency. The crossing (generation 12) and the subsequent resistance takeover are both visible; this transient is the best case the grid offers, and it is still not a viable release profile.

generation	wild-type W	drive D	resistant R
0	0.9900	0.0100	0.0000
3	0.9263	0.0737	0.0000
6	0.6160	0.3840	0.0000
9	0.1752	0.8248	0.0000
12	0.0345	0.9602	0.0053
15	0.0080	0.3826	0.6094
18	0.0038	0.1818	0.8144
21	0.0024	0.1188	0.8787
24	0.0018	0.0882	0.9100
27	0.0014	0.0701	0.9285
30	0.0012	0.0581	0.9407
33	0.0010	0.0497	0.9493

36	0.0009	0.0434	0.9558
39	0.0008	0.0385	0.9608
42	0.0007	0.0346	0.9647
45	0.0006	0.0314	0.9680
48	0.0006	0.0287	0.9707
51	0.0005	0.0265	0.9730
54	0.0005	0.0246	0.9749
57	0.0005	0.0229	0.9766
60	0.0004	0.0215	0.9781

The drive peaks at generation 12 ($D = 0.960$) and is overtaken by resistance within two generations; by generation 20 $R = 0.84$ and the drive is in terminal decline. The table makes concrete what the grid aggregates: under full sterility there is no stable internal equilibrium, only a race between homing and resistance that resistance always wins at these parameters.

Appendix G. Computational cost

All steps run within the program's 1 GB RAM / 2 CPU sandbox. Scanning is linear in region length: 110,001 windows per strand, constant work per window. CNN scoring is one batched forward pass over 9,328 sites. Off-target screening is the only superlinear step: the 5-mer inverted index over the 110 kb region occupies $O(L)$ memory for region length L , and each query touches only the buckets of its four 5-mer blocks, so the observed cost is a small constant multiple of the number of true plus incidental block collisions rather than the $O(L)$ windows an exhaustive scan compares against. In practice the top-150 off-target screen completes in seconds on the sandbox hardware; the same index design scales to the ~280 Mb genome by block-sharding the index, which is why the genome-wide extension (Limitation 2) is an engineering task, not a methods problem. The 60-generation deterministic grid (16 cells) and the Wright-Fisher replicates (5 seeds x 60 generations x $N = 10,000$) are negligible by comparison.

Measured wall-clock on the sandbox (single process):

step	workload	time
locus scan + CNN score	10,936 sites, 9,328 scored	seconds
region off-target screen	top-150 guides x 110 kb	seconds
genome-wide enumeration	30 guides x 245.46 Mb, ≤ 3 mm	76.75 s
drive grid	16 cells x 60 generations	< 1 s
Wright-Fisher	5 seeds x 60 generations x $N = 10,000$	seconds
full test suite	36 tests	seconds
paper build	markdown to Times PDF	seconds

Appendix H. Negative results and preserved failures

In keeping with the program's honest-verdict discipline, we record what did not work or did not hold. (1) The efficacy-first design intuition fails: the highest-efficacy site in the locus is not the best design once constraint is scored, and the field's validated site ranks only 94th of 150 on efficacy alone. (2) The drive fails in 15 of 16 simulated parameter cells; this study produces no viable release-ready design under full sterility, and we regard that as the correct scientific output rather than a shortcoming. (3) The current RefSeq annotation does not reproduce the published exon-junction description of the Kyrou target; the discrepancy was pinned by a test, reported rather than smoothed over, and is now resolved as a gene-model difference between annotation releases (Appendix I). (4) Human-cell-trained efficacy scores do not establish mosquito efficacy; the candidate lists are hypotheses with quantified priors, and we explicitly do not claim them as validated guides.

Appendix I. The Kyrou annotation discrepancy, resolved against AgamP4

The published target is described as spanning the intron-4/exon-5 boundary of the female-specific *dsx* transcript under the AgamP4 (VectorBase) annotation, while the current RefSeq exon model on the AgamP5 assembly places the site inside an intron, about 0.5 kb from the 135-bp female-specific exon. Earlier sections pinned the discrepancy without resolving it; the queued re-annotation is now done.

Method (scripts/agamp4_reannotation.py, hermetic; evidence archived under data/agamp4/). We fetched the AgamP4 chromosome 2R annotation and sequence from Ensembl Metazoa's current release (*Anopheles gambiae*.AgamP4.63.chromosome.2R.gff3.gz, which carries the VectorBase community genebuild, and the matching soft-masked 2R DNA), extracted the full AGAP004050 model, and located the Kyrou protospacer by exact sequence match.

Findings. (1) Under AgamP4, *dsx* (2R:48,703,664-48,788,460, minus strand) has two transcripts: AGAP004050-RA (6 exons) and AGAP004050-RB (7 exons), the second distinguished by an extra 1,692-bp exon (2R:48,712,957-48,714,648) whose terminal 92 bp are coding. (2) The Kyrou protospacer matches exactly once in AgamP4, at 2R:48,714,640-48,714,659 (minus strand), and 9 of its 20 bases (48,714,640-48,714,648) lie inside that extra exon's coding sequence: the guide spans the exon's splice boundary with 9 coding bp plus 11 intronic bp. The 2018 description is therefore correct under the annotation it used: resolution (i) of the earlier Appendix is confirmed - the RefSeq/AgamP5 model simply lacks the RB-specific exon, which is why the current record calls the site intronic. (3) The 135-bp female-specific exon exists identically in both annotations (AgamP4 2R:48,715,161-48,715,295; AgamP5 NC_064601.1:47,622,697-47,622,831). (4) Assembly sensitivity of the named leads: mapped onto AgamP4, *dsx*-v3-2's protospacer carries exactly one mismatch, at guide position 20 - the base immediately adjacent to the PAM, the most cleavage-critical position - and *dsx*-v3-1's carries three. The AgamP4 and AgamP5 references differ at real sequence positions inside the target region, so a guide that is exact against one

assembly may be a seed-mismatch guide against the other. This is a concrete, quantified demonstration of Limitation 2 and the strongest internal argument for the Ag1000G population-level enumeration of Section 4.7: specificity and efficacy must both be verified against population sequence, not a single reference.

The discrepancy is thus closed: both annotations are internally consistent, they disagree in gene model rather than in sequence truth, the 2018 authors described their site correctly, and our constraint scoring - which rewarded the region under either annotation - is unaffected in its ranking conclusions. All findings are pinned by tests/test_agamp4.py (3 tests), including the 9-bp coding overlap and the leads' mismatch positions, so any future annotation update that changes these facts fails the suite.

Appendix J. Mathematical derivations

J1. Pigeonhole completeness of the candidate filter. Claim: if a 20-mer window w differs from protospacer p in at most 3 positions, then at least one of the four contiguous 5-mer blocks of w equals the corresponding block of p exactly. Proof: partition the $L = 20$ positions into four disjoint blocks of 5. Suppose every block contained at least one mismatch; then the total number of mismatches is at least 4, contradicting the hypothesis of at most 3. Hence some block is exact. The filter therefore enumerates a superset of the true ≤ 3 -mismatch set, and exact verification on the superset yields exact counts. The same argument applied to the reverse complement covers the opposite strand, doubling the block set to 8.

J2. Block-overlap coverage. Claim: streaming the reference in blocks of B bases with consecutive-block overlap of $L - 1 = 19$ bases places every 20-mer window wholly inside at least one block. Proof: block k starts at position $k(B - 19)$. A window starting at s lies wholly in block k if $k(B - 19) \leq s$ and $s + 19 \leq k(B - 19) + B - 1$. The second condition rearranges to $s \leq k(B - 19) + B - 20 = (k + 1)(B - 19) - 1$. So any s with $k(B - 19) \leq s \leq (k + 1)(B - 19) - 1$ is covered, and these intervals tile the non-negative integers contiguously. Every window start is covered.

J3. Expected near-cognate count under a uniform random genome. For a fixed protospacer of length L and a random position with i.i.d. uniform bases, the number of mismatches D is Binomial($L, 3/4$): $P(D = i) = C(L, i) (3/4)^i (1/4)^{L-i}$. Over G window positions on both strands, the expected number of sites with at most d mismatches is $E[N] = 2G * \sum_{i=0}^d C(L, i) 3^i / 4^L$. For $L = 20$, $d = 3$, the per-position probability is $\sum_{i=0}^3 C(20, i) 3^i / 4^{20} = (1 + 60 + 1710 + 20520) / 1.0995e12 = 2.027e-8$, giving $E[N] = 9.95$ for $G = 245.46e6$ - versus observed genome-wide 3-mismatch-plus-closer counts of 4-45 across our candidates, consistent with order-of-magnitude uniformity outside repetitive protospacers (the two >300 -count guides violate it, flagging low complexity, section 3.9).

J4. k -mer index construction cost. Building the 5-mer index over a block of B bases is $\Theta(B)$ time and $\Theta(B)$ space: each of $B - 4$ positions inserts one entry. Candidate generation for one

guide costs $O(\sum \text{over its 8 blocks of } |\text{index}(\text{block})|)$, and verification costs $O(L)$ per candidate, so per-guide per-block cost is $\Theta(B)$ amortized index plus $O(cL)$ for c candidates.

J5. CNN layer equations. The efficacy model computes $h^{(1)} = \text{ReLU}(\text{conv1D}(x_{\text{onehot}}, W1) + b1)$ with kernel width 5 over the 30-mer one-hot input (30×4), followed by global pooling $g = \text{mean}_t h^{(1)}_t$, and a linear head $\hat{y} = w^T g + b$. Training minimizes the mean squared error $\text{MSE} = (1/n) \sum_j (\hat{y}_j - y_j)^2$ over the 4,248-example training split with Adam (15 epochs, seed 0).

J6. Spearman rank correlation. With $R(x_i)$, $R(y_i)$ the ranks of paired scores, $\rho = 1 - 6 \sum_i d_i^2 / (n(n^2 - 1))$, $d_i = R(x_i) - R(y_i)$. Our held-out value 0.556 (cross-dataset 0.609) uses this form on the 1,062-example test split, computed identically for CNN and ridge baselines.

J7. Ridge baseline normal equations. The sequence-agnostic baseline solves $w = (X^T X + \lambda I)^{-1} X^T y$ on the same one-hot features, $\lambda = 1.0$; its held-out Spearman 0.463 and cross-dataset 0.444 quantify the CNN's margin over the linear rule on identical splits.

J8. Deterministic drive recursion. With wild-type w , drive d and resistance r frequencies summing to 1, the per-generation update under homing rate h , resistance-generation rate e and full conversion ($c_{\text{hom}} = 1$) is: $d' = d + h w (1 - e) N_d$, $r' = r + h w e N_r$, $w' = w - h w N_w$, where the N terms normalize for genotype frequencies under random mating; the grid varies h in $\{0.90, 0.95, 0.99, 1.00\}$ and e in $\{0.001, 0.01, 0.1, 0.5\}$. The recursions are iterated 60 generations and the implementation is pinned against closed-form spot values by `test_drive_sim.py`.

J9. Resistance-dominance threshold. Resistance dominates when $r_t > d_t$. Dividing the recursions, r/d evolves approximately as $(r + h w e)/(d + h w (1 - e))$; for $e > 1/2$ the ratio grows every generation regardless of h , and for small e the drive transiently leads before resistance accumulates - the grid's uniform resistance dominance by generation 60 is the discrete analogue of the continuous result that any positive e eventually dominates when resistant genotypes are fertile.

J10. CFD-style specificity penalty. Empirical Cas9 cutting at mismatched sites decays sharply with mismatch count and seed proximity; we encode the decay as a geometric penalty weight per mismatch class, $w_i = \gamma^i$ with $\gamma = 0.1$ in log-space units, operationalized as 0.50 per 2-mismatch site and 0.05 per 3-mismatch site - a 10:1 ratio matching the order-of-magnitude activity drop reported for the third mismatch class in CFD-calibrated surveys.

J11. v3 as constrained optimization. The design problem is $\text{argmax over } g \text{ of } E(g) + C(g) - \sum_i w_i \text{mm}_i(g)$ subject to $\text{mm}_0(g) + \text{mm}_1(g) = 0$, where E is CNN efficacy, C the constraint bonus, and the mm_i exact genome-wide counts. The hard constraint implements the safety filter; the linear penalty trades efficacy against residual near-cognate burden.

J12. Wright-Fisher stochastic update. Each generation samples offspring genotype counts from a multinomial with probabilities given by the deterministic frequencies: $(n_w, n_d, n_r) \sim \text{Multinomial}(N, (w', d', r'))$, then $(w, d, r) = (n_w, n_d, n_r) / N$. Five seeds at $N = 10,000$ bracket

the deterministic trajectory (Figure 5), quantifying drift around the recursion of J8.

J13. NGG site density under composition. For i.i.d. bases with $P(G) = g$, a random 3-mer is an NGG PAM with probability g^2 ; both strands of a region of length B therefore contain $E[S] = 2(B - 2) g^2$ sites. Under the region's observed GC fraction 0.4626 with $g = c = 0.2313$, $E[S] = 2 \times 109,999 \times 0.0535 = 11,770$; the observed count is 10,936 (5,091 plus, 5,845 minus), a 7% deficit attributable to local composition fluctuation and dinucleotide structure. Under the uniform prior $g = 1/4$ the expectation is 13,750. The asymmetry between strands (5,091 vs 5,845) reflects the G-skew of this region, not a strand bias of the scanner, and the counts are pinned by the scan tests.

J14. Information-theoretic uniqueness. A specified 20-mer carries 40 bits ($\log_2 4^{20}$). A genome of G bases presents $2G$ overlapping windows, so the expected number of exact chance matches is $2G / 4^{20} = 4.5e-4$ for $G = 245.46e6$. Exact uniqueness is therefore the prior expectation for every specific 20-mer, and observing any exact second copy is evidence of duplication or repeat biology, not chance. This is the information-theoretic complement of J3 and explains why all 30 candidates and the control returned $mm0 = 0$ genome-wide.

J15. Rule-of-three bound for population variants. Reference-genome enumeration cannot see segregating variation. If none of n independent chromosomes sampled from the population carried a ≤ 1 -mismatch near-cognate site, the upper 95% confidence bound on the population frequency of such a site is $3/n$ (Hanley and Lippman-Hand 1983). The Ag1000G measurement of Section 4.7 carries this out at the three candidate windows (a 9,386-chromosome cohort), replacing the bound with an enumeration there; for sites outside those windows, $3/n$ with the n of available assemblies remains the honest specificity statement.

J16. Multiplex escape probability. If a drive carries k guides cutting the same essential site and resistance-conferring indels arise independently at rate e per cut per site, a chromosome is drive-resistant only when all k sites are resistant: $P(\text{escape per homing event}) = e^k$ under independence. Independence is an idealization - homology-driven repair of one cut can template the other, so the achievable floor is correlated - but the bound is the design target: with $e = 0.01$ per site, two guides give $1e-4$, three give $1e-6$ per event, pushing expected time-to-resistance beyond any cage or release horizon in Section 3.4's grid. This is the quantitative form of the multiplex strategy of Section 4.3, and it is why the v3 shortlist's mutual proximity (both leads inside 1.3 kb) matters: a single construct can carry both.

J17. GC-content window under composition. The GC count of a 20-mer is $\text{Binomial}(20, p)$ with p the local GC fraction, approximated by $\text{Normal}(20p, 20p(1-p))$ for $20p(1-p) > 9$. The design filter $0.40 \leq \text{GC} \leq 0.60$ (8-12 G+C bases) corresponds to z in $[-1.83, +1.83]$ at $p = 0.5$, so the filter keeps the central 93% of the uniform prior; its bite comes from excluding extreme-composition guides whose kinetics (very high GC) or R-loop stability (very low GC) are pathological, not from reshaping the bulk.

J18. Drive fixation probability. A homing allele transmitted to fraction $(1 + h_t)/2$ of a heterozygote's offspring gains effective selection coefficient $s = h_t/2$ per generation against the wild-type when fitness costs are absent. Under the diffusion approximation its fixation probability from initial frequency p_0 is $P_{\text{fix}} = (1 - e^{-2 N_e s p_0}) / (1 - e^{-4 N_e s})$ in the haploid-scaled convention; for large $N_e s$ the numerator saturates at 1 and $P_{\text{fix}} \rightarrow 1 - e^{-4 N_e s p_0}$, so even a single introduction fixes with probability about $2s = h_t$ when conversion is efficient and resistance is absent. The grid of Section 3.4 sits in the regime where this numerator is irrelevant because resistance (J8, J9) dominates first.

J19. Wright-Fisher drift magnitude. Conditional on current drive frequency d , the next generation's frequency has variance $d(1 - d)/(2N)$ per generation. At the Section 3.5 population size $N = 10,000$, one generation of drift at $d = 0.5$ has standard deviation $\sqrt{0.25/20,000} = 0.0035$, and 60 generations of accumulated drift remain below 0.03 in frequency units - which is why the five stochastic replicates bracket the deterministic trajectory tightly (Figure 5) and why the resistance-dominance conclusion is a selection result, not a drift artifact.

J20. CFD as a product of position-specific penalties. The Cutting Frequency Determination score of Doench et al. (2016) assigns each (position, reference base, variant base) triple an empirical penalty $m(p, a, b)$ in $(0, 1]$ and scores a mismatched site as $\text{CFD} = \text{product over mismatched positions } p \text{ of } m(p, a_p, b_p)$. Multiplicativity assumes log-independent contributions. Our geometric penalty (J10) is the rank-1 approximation $m(p, a, b) = \gamma$ independent of position and identity; it is exact when penalties are uniform and is used here only to rank candidates whose exact low-mismatch counts are already known, so the approximation cannot hide a dangerous site - it can only mis-order two safe ones.

J21. Why specificity is a tail problem. For two unrelated 20-mers the Hamming distance D is $\text{Binomial}(20, 3/4)$, with mean 15 and standard deviation $\sqrt{20 \times 0.75 \times 0.25} = 1.94$. A site at 3 mismatches is therefore a 6.2-standard-deviation event in the left tail. This explains the empirical shape of Section 3.9.1: the 3-mismatch class carries nearly all near-cognate mass, the 2-mismatch class is rare, and the 1-mismatch class is nearly empty - each step inward costs a factor of roughly 30 in probability (from the binomial ratio $P(D = i - 1)/P(D = i) = i/(3(21 - i))$, which is $1/57$ at $i = 3$).

J22. Order statistics of the efficacy distribution. If the 9,328 gene-body scores were i.i.d. normal with the observed mean 0.763 and standard deviation 0.178, the expected maximum would be about $\text{mean} + \text{sd} \times \sqrt{2 \ln n} = 0.763 + 0.178 \times 4.27 = 1.523$; the observed maximum is 1.345, below the null expectation. There is no magical site: the top of the efficacy ranking is statistically unremarkable, which is the quantitative justification for not designing on efficacy alone and for letting constraint (v2) and specificity (v3) drive the final ranking.

J23. Multiplex escape under correlated repair. Independence (J16) is the optimistic bound. Model the repair context of co-located cuts as shared with correlation ρ : then $P(\text{all } k \text{ sites resistant per})$

event) is bounded between e^k (independence) and e (identical outcomes), interpolable as $e^k + \rho(e - e^k)$. The design responses follow directly: prefer guides whose protospacers differ in sequence context (reducing shared repair biases), and prefer spatial separation that still fits one construct - the v3 tier's 3.8 kb cluster (Section 3.13.4) satisfies the second, and sequence diversity across the tier satisfies the first.

J24. Early-growth closed form. While the drive is rare (d small, resistance absent), heterozygote frequency is approximately $2d$ and each generation converts fraction h of wild-type alleles in those heterozygotes, giving $d_{t+1} \sim (1 + h) d_t$. The drive grows geometrically with per-generation factor $(1 + h)$: from a 1% introduction, reaching 50% takes $\ln(50) / \ln(1 + h) = 5.6$ generations at $h = 0.99$, and the decelerating tail (heterozygote frequency falls as d approaches 1) places the 95% crossing a few generations later - the grid's observed crossing at generation 12 (Section 3.4) matches this closed-form accounting. The same formula shows why h barely matters to the final outcome: $\ln(1.99) / \ln(1.80) = 1.17$, so even a large h improvement shortens the growth phase by under 20%, while e controls whether the drive survives past it.

J25. Sampling error of the Spearman comparison. Under independence between predicted and measured ranks, the Spearman estimator over n guides has variance about $1/(n - 1)$; for the $n = 1,062$ held-out split the standard error is 0.031. The CNN's same-split advantage over ridge (+0.093) is three such standard errors, and the paired nature of the comparison (both models rank the same guides, so their errors correlate) makes the effective test stronger, not weaker; the cross-dataset advantage (+0.166 over $n = 2,144$, standard error 0.022) is larger still. The comparison is statistically comfortable; the caveat that limits it is distributional (human-cell data, random split), not statistical power (Appendix N).

J26. Expected filter pass count under the random model. From J3, the per-position probability of a site at ≤ 1 mismatch is $(1 + 60)/4^{20} = 5.55e-11$, giving a per-guide expectation $E[N_{\{\leq 1\}}] = 2G \times 5.55e-11 = 0.0272$ over $G = 245.46e6$. Under the Poisson limit, a guide passes the hard filter with probability $e^{-0.0272} = 0.973$, so 30 guides yield 29.2 expected passes; the observed count is 28. The small deficit (2 failures against 0.8 expected) is within binomial sampling error (standard deviation $\sqrt{30 \times 0.973 \times 0.027} = 0.89...$ at face value 1.2 failures expected from the deficit alone is not anomalous) and, more informatively, the failures are not random draws: they are the candidates whose protospacers happen to have near-relatives in the genome, which is exactly what the filter exists to catch. The agreement between the random-model expectation and the observed pass rate is a cross-check on the enumerator, because a bug that dropped true hits would inflate the pass rate above the expectation.

J27. The observed 3-mismatch excess over the random model. J3 predicts $E[N_{\{\leq 3\}}] = 9.97$ near-cognate sites per guide under the uniform model; the observed genome-wide mean over the 30 candidates is $1,469/30 = 49.0$, a 4.9-fold excess. Two mechanisms produce it, both visible in the data. First, the two repetitive, low-complexity guides contribute over 600 sites between them - under the random model each would expect about 10, so they carry a roughly 30-fold individual

excess driven by sequence repetitiveness, the known failure mode of the i.i.d. assumption. Second, even excluding them, the remaining 28 guides average about 29 sites each, a residual ~3-fold excess consistent with real genomes being compositionally structured (CpG suppression, microsatellite content, segmental duplication) rather than i.i.d. uniform. The practical consequence is a calibration warning: per-guide specificity cannot be estimated from protospacer-uniform formulas; it must be enumerated, which is the operational reason Section 2.9 exists. The random model remains useful as the cross-check of J26, where the ≤ 1 -mismatch class is so rare that repeat structure does not reach it.

Appendix K. Genome-wide v3 table (30 candidates + published control)

Identity and score columns; mismatch counts in Table K2.

rank_v3	protospacer	str	pos (NC_064601.1)	efficacy	v3
1	TGGGCAGTATGCGTTAGGGT	-	47619040	0.925	1.2248
2	CATTAAGACCTACGAAGCGC	-	47620304	0.900	1.2004
3	GAAGCGAGCCCAATGGCTGT	-	47622796	0.937	1.1867
4	CCCAGCCGCAGCATATGGG	-	47692039	0.750	1.1001
5	TGCAGTGAAACCCGCGTGGA	-	47695204	1.285	1.0349
6	TTACCAGGTAGGAGATTTAC	+	47655247	0.682	1.0317
7	CTAGCTCGTCGTCTGAAATG	+	47622820	0.952	1.0023
8	ACTAGCTCGTCGTCTGAAAT	+	47622819	0.749	0.9992
9	GCGGTACACTGCACTGTCCG	+	47619081	1.209	0.8589
10	AGCTAGTGAAGCGAGCCCAA	-	47622803	1.101	0.8011
11	GAACCTGTAAATCTCCTACC	-	47655250	0.901	0.7013
12	TCGAGGAAGCGATGGTGCTA	+	47693451	0.841	0.4906
13	GGTAGGAGATTTACAGGTTC	+	47655253	0.678	0.4784
14	CACTAGCTCGTCGTCTGAAA	+	47622818	0.698	0.4481
15	TATGCTGCGGCTCGGGAACG	+	47692046	0.698	0.4475
16	ACCTCTCCGAACGTTTGCAT	-	47700884	0.702	0.3022
17	CCGTTGCAGTGAAACCCGCG	-	47695208	1.203	0.2527
18	TACTTACCTCCCATATGCTG	+	47692033	0.958	-0.3421
19	TGCGGCTCGGGAACGTGGTG	+	47692051	0.880	-0.4196
20	GTCGAGGAAGCGATGGTGCT	+	47693450	0.676	-0.4235
21	GCACACCAGCGGATCGACGA	-	47622696	0.994	-0.8559
22	ACACTTACCTTTTTGTACAG	+	47619062	0.935	-0.9647
23	ACGACGGATACAGACTGGA	+	47619412	1.345	-1.2553
24	GTGCGAAGCGCACGAAACAC	+	47700905	0.711	-2.0392

25	ATGTACAAAAAGCACACCAG	-	47622707	1.072	-3.0284
26	TGTGTCGTTTACCTTTTTGG	+	47620325	0.865	-4.435
27	GCTTACCTTCGTCGATCCGC	+	47622691	0.830	-23.2198
28	GCGGATCGACGAAGGTAAGC	-	47622688	0.756	-23.2937
	CAGTGTACCGCTGTACAAAA	-	47619069	0.973	
	ACGGTGGCGCGCGAGCGAGA	-	47696369	0.672	
control	GTTTAACACAGGTCAAGCGG	-	47622174	1.137	0.837

Table K2. Exact genome-wide off-target counts by mismatch class (AgamP5, 246 Mb) with the hard-filter outcome.

protospacer	mm0	mm1	mm2	mm3	filter	constraint
TGGGCAGTATGCGTTAGGGT	0	0	0	4	pass	intronic/intergenic,splice_proximal(d=8)
CATTAAGACCTACGAAGCGC	0	0	0	9	pass	exonic(23/23),splice_proximal(d=10)
GAAGCGAGCCCAATGGCTGT	0	0	0	10	pass	exonic(23/23),splice_proximal(d=13)
CCCGAGCCGCAGCATATGGG	0	0	0	8	pass	exonic(22/23),splice_proximal(d=0)
TGCAGTGAAACCCGCGTGGA	0	0	0	10	pass	exonic(23/23)
TTACCAGGTAGGAGATTAC	0	0	0	8	pass	exonic(19/23),splice_proximal(d=0)
CTAGCTCGTCGTCTGAAATG	0	0	0	14	pass	exonic(12/23),splice_proximal(d=0)
ACTAGCTCGTCGTCTGAAAT	0	0	0	10	pass	exonic(13/23),splice_proximal(d=0)
GCGGTACACTGCACTGTCCG	0	0	1	12	pass	exonic(23/23),splice_proximal(d=11)
AGCTAGTGAAGCGAGCCCAA	0	0	1	11	pass	exonic(23/23),splice_proximal(d=6)
GAACCTGTAAATCTCCTACC	0	0	1	9	pass	exonic(22/23),splice_proximal(d=0)
TCGAGGAAGCGATGGTGCTA	0	0	1	12	pass	exonic(23/23),splice_proximal(d=10)
GGTAGGAGATTTACAGGTTT	0	0	1	9	pass	exonic(23/23),splice_proximal(d=2)
CACTAGCTCGTCGTCTGAAA	0	0	1	10	pass	exonic(14/23),splice_proximal(d=0)
TATGCTGCGGCTCGGGAACG	0	0	0	20	pass	exonic(23/23),splice_proximal(d=6)
ACCTCTCCGAACGTTTGCAT	0	0	1	13	pass	exonic(23/23),splice_proximal(d=14)
CCGTTGCAGTGAAACCCGCG	0	0	1	14	pass	exonic(23/23)
TACTTACCTCCCATATGCTG	0	0	2	21	pass	exonic(16/23),splice_proximal(d=0)
TGCGGCTCGGGAACGTGGTG	0	0	2	21	pass	exonic(23/23),splice_proximal(d=11)
GTCGAGGAAGCGATGGTGCT	0	0	2	17	pass	exonic(23/23),splice_proximal(d=11)
GCACACCAGCGGATCGACGA	0	0	3	22	pass	exonic(22/23),splice_proximal(d=0)
AACTTACCTTTTTGTACAG	0	0	3	23	pass	exonic(15/23),splice_proximal(d=0)
ACGGACGGATACAGACTGGA	0	0	1	47	pass	exonic(23/23)
GTGCGAAGCGCACGAAACAC	0	0	5	20	pass	exonic(16/23),splice_proximal(d=0)
ATGTACAAAAAGCACACCAG	0	0	4	57	pass	exonic(23/23),splice_proximal(d=10)
TGTGTCGTTTACCTTTTTGG	0	0	6	61	pass	exonic(12/23),splice_proximal(d=0)

GCTTACCTTCGTCGATCCGC	0	0	4	456	pass	exonic(17/23),splice_proximal(d=0)
GCGGATCGACGAAGGTAAGC	0	0	4	456	pass	exonic(14/23),splice_proximal(d=0)
CAGTGTACCGCTGTACAAAA	0	1	5	39	FAIL	exonic(22/23),splice_proximal(d=0)
ACGGTGGCGCGCGAGCGAGA	0	1	2	46	FAIL	exonic(23/23),splice_proximal(d=2)
GTTTAACACAGGTCAAGCGG	0	0	0	6	pass	published guide (Kyrou 2018)

Appendix L. Kyrou 2018 guide - complete genome-wide audit

Guide: GTTTAACACAGGTCAAGCGG (5'→3'). Reference: AgamP5 GCF_943734735.2. Zero sites at 0, 1 or 2 mismatches. Complete list of 3-mismatch sites:

chromosome	position (0-based)	strand	mismatches
NC_064601.1	41285162	-	3
OX030909.1	6534277	-	3
OX030909.1	14585141	-	3
OX030908.1	7690618	-	3
OX030908.1	81101940	+	3
OX030908.1	91899450	+	3

On-target locus (NC_064601.1:~47,622,174, minus strand) excluded from the count as the designed cut site.

Appendix M. Enumerator benchmark against published Cas-OFFinder numbers

Published Cas-OFFinder performance (Bae, Park and Kim 2014; Intel i7-3770K CPU, AMD Radeon HD 7870 GPU):

experiment	genome size	mismatches	time
1,000 targets, CPU	human 3.01 Gb	unspecified	60.0 s
1,000 targets, GPU	human 3.01 Gb	unspecified	3.0 s
100 targets, GPU	human 3.01 Gb	1	76.4 +/- 2.0 s
100 targets, GPU	human 3.01 Gb	5	79.9 +/- 1.6 s
100 targets, GPU	human 3.01 Gb	10	114.4 +/- 3.0 s
100 targets, GPU	mouse 2.65 Gb	5	62.6 +/- 2.4 s
100 targets, GPU	zebrafish 1.32 Gb	5	37.7 +/- 3.5 s
100 targets, GPU	Arabidopsis 116 Mb	5	4.8 +/- 0.8 s

Normalization. Throughput per guide-gigabase is the comparable unit: seconds per (number of guides x genome size in Gb). Our enumerator: 76.75 s / (30 x 0.2459) = 10.4 s per guide-Gb on one

CPU thread. Cas-OFFinder CPU: $60.0 / (1,000 \times 3.01) = 0.0199$ s per guide-Gb. Cas-OFFinder GPU: $3.0 / (1,000 \times 3.01) = 0.000997$ s per guide-Gb. The per-100-target rows are consistent with the amortized rows once fixed launch cost is removed.

enumerator	hardware	s per guide-Gb	speedup vs ours
this work	1 CPU thread (sandbox)	10.4	1x
Cas-OFFinder	i7-3770K CPU	0.0199	523x
Cas-OFFinder	Radeon HD 7870 GPU	0.000997	10,431x

(Figure 10 plots the same comparison on a log scale.)

Verdict (honest negative). On raw enumeration throughput our scanner loses to published Cas-OFFinder by roughly 5×10^2 on CPU and 10^4 on GPU, and the gap would grow with mismatch budget because their bit-parallel comparison cost is nearly flat in mismatches. Protocol differences (2014 hardware, their larger PAM alphabet, our NumPy interpreter overhead) shift the constant, not the order of magnitude. We do not claim a speed record. What the deficit buys is exactness with a one-page completeness proof (J1-J2), zero external dependencies, and per-guide latency low enough (2.56 s) to sit inside the design loop at insect-genome scale. If this pipeline moves to production libraries of thousands of guides, the correct move is to swap the enumerator for a GPU kernel with the same exact-verification contract; the ranking, simulation and certificate layers are unaffected.

Appendix N. Efficacy-model protocol note (why we do not claim to beat Rule Set 2)

Published efficacy numbers come from incompatible protocols. Rule Set 2 / Azimuth (Doench et al. 2016) reports performance under gene-level holdout schemes on its tiling data; CRISPOR's independent evaluation (Haeussler et al. 2016) reports dataset-dependent Spearman values for the same model; our 0.556 comes from a single random 80/20 split of the FC+RES data. Random splits leak sequence similarity across the partition and inflate absolute numbers relative to gene-level holdout. Comparing our 0.556 to any published Azimuth number would therefore be a protocol error, not a result. The honest comparisons we can stand behind: (i) same split, same metric - CNN 0.556 vs ridge 0.463; (ii) same models, untouched external data - 0.609 vs 0.444 on Doench V1. A protocol-matched head-to-head (retraining gradient-boosted Rule Set 2 features on our exact split) is queued as future work and is the only form the claim "the CNN beats Rule Set 2" is allowed to take in this paper.

Appendix O. Coordinate conventions and notation

Three coordinate systems appear in this paper and its result files:

1. Region FASTA (data/dsx_locus_region.fasta): fetched as NC_064601.1:47600000-47710000 under NCBI's 1-based inclusive convention, so region offset i (0-based) equals genomic 0-based coordinate $47,599,999 + i$. 2. Candidate CSVs (ranked_designs.csv): *genomic_pos* is the 1-based coordinate of the first PAM base. For minus-strand candidates the protospacer occupies $genomic_pos + 3 \dots genomic_pos + 22$ (1-based inclusive). Example: dsx-v3-1 at *genomic_pos* 47,619,040 has PAM AGG at 47,619,040-42 and protospacer at 47,619,043-62. 3. Genome-scan hit lists (results/genome_offtargets_.json, Section 3.13 tables): 0-based start of the matched 20-mer window on the plus strand; strand '-' means the reverse complement of the window aligns to the guide.

Notation: $L = 20$ (protospacer length); G = genome size in bases; mm_i = count of genomic sites at exactly i mismatches; e = resistance-generation rate per cut; h = homing rate; c_{hom} = homing conversion fraction; d = distance to the nearest splice junction; $v3$ = composite score of Section 2.10; CFD = cutting frequency determination score (J20).

Appendix P. External tools and resources

Counting rule. A tool is counted only if it was actually used in the work reported here, with its role named. Tools that are planned but not yet run are listed separately at the end and are not counted. This table is the honest inventory; it is not padded to reach any target.

#	tool / resource	type	what it was used for
1	NCBI E-utilities (efetch, esearch, esummary, elink)	web API	archived the dsx locus FASTA (NC_064601.1:47,600,000-47,710,000), the RefSeq gene table, the NCBI Gene 1270904 record, and the nine dsx transcript sequences (Appendix A)
2	NCBI RefSeq assembly AgamP5 (NC_064601.1 and three further molecules)	database	reference genome for target scanning, genome-wide off-target enumeration, and the Kyrou audit (2.9, 3.8, 3.9, 3.10)
3	Ensembl Metazoa FTP (AgamP4.63 GFF3; AgamP4 dna_sm FASTA; VectorBase genebuild)	database	independent annotation that resolves the Kyrou annotation discrepancy (Appendix I) and the reference for the population windows (4.7)
4	MalariaGEN Ag1000G phase 3, Sanger VCF mirror	database	per-sample variant calls at the candidate windows across the wild population (4.7)
5	Europe PMC API	web API	full text of Kyrou et al. 2018 (PMC6871539), source of the positive-control gRNA sequence
6	Microsoft Research Azimuth data mirror	dataset repository	Doench 2016 FC+RES (5,310 guides) training data and the untouched Doench V1 (2,144 guides) external check (2.3)

7	WHO World Malaria Report 2024	report	2023 burden figures framing the problem (263M cases, 597,000 deaths; 1.1)
8	Cas-OFFinder (Bae, Park, Kim 2014)	published tool	head-to-head enumeration benchmark; an honest loss, preserved (Appendix M)
9	Rule Set 2 / Azimuth (Doench et al. 2016)	published model	efficacy-model protocol comparison; we do not claim to beat it (Appendix N)
10	Kyrou et al. 2018 dsx drive reagent	published reagent	positive control, constraint target, and genome-wide audit subject (3.1, 3.10)
11	NumPy	package	k-mer index, off-target counting, Wright-Fisher simulation, numerics throughout
12	PyTorch	package	CNN efficacy model: architecture, training, inference (2.3)
13	matplotlib	package	figures 1-10
14	reportlab	package	Times-font PDF rendering of this paper
15	pysam / htlib (tabix)	package	indexed remote queries of the Ag1000G per-sample VCFs (4.7)
16	pytest	package	the 77-test hermetic verification suite (Appendix C)
17	git / GitHub	platform	versioned public archive of code, data, and results
18	Biopython	package	independent re-derivation of the three AF-scan window sequences and PAMs from the archived AgamP4 FASTA; all three pass (results/window_verification_biopython.json)
19	ViennaRNA (RNAfold)	package	spacer secondary-structure screen on the named candidates; no hairpin flags (results/guide_fold_primers.json)
20	Primer3	package	validation-amplicon primer pairs spanning each candidate window for the staged plan of 4.6 (results/guide_fold_primers.json)
21	UniProtKB (REST API)	database	sex-specific dsx isoform corroboration: female-specific (A0A1U7F5R8, 241 aa) and male-specific (Q58QC2, 283 aa) proteins archived at data/uniprot_dsx_anoga.tsv; the female-exon target strategy rests on this isoform biology
22	minimap2 2.28	aligner	orthogonal cross-assembly check: AgamP4 dsx region maps to AgamP5 2RL in 10 collinear blocks, all three guide windows within 7-12 bp of the aligner-free coordinates (results/cross_assembly_minimap2.json; 4.10)
23	MMseqs2	package	redundancy audit of the population near-cognate inventory: 43 allele 23-mers cluster to 27 groups at 87% identity, all multi-member clusters intra-guide (results/allele_clusters_mmseqs2.json; 4.8)

24	NCBI BLAST+ 2.17.0	aligner suite	orthogonal off-target confirmation: blastn-short of the three 23-mer guide units against a local AgamP5 DB finds exactly one full-length hit per guide (its own site) and zero near full-length hits (results/offtarget_blast.json; 4.10)
25	seqkit 2.10.0	package	independent molecule-stat audit: four archived AgamP5 molecules total 245,459,960 bp, matching Section 2.1.1 and correcting stale '245.9 Mb' mentions (results/genome_stats_seqkit.json)
26	Bowtie 1.3.1	aligner	fourth off-target confirmation: -v 3 full-length <=3-mismatch search reports exactly the three on-target sites genome-wide, zero secondary (results/offtarget_bowtie.json; 4.10)
27	ViennaRNA 2.7.2	RNA folding library	spacer-structure QC: dsx-v3-3 folds at -4.70 kcal/mol, z=-2.11 vs a 500-mer 2R null - a documented efficacy-confound flag on the third lead; v3-1, v3-2 and the Kyrou control are unremarkable (results/guide_structure_vienna.json; 4.10)
28	CHOPCHOP v3 (web)	curated guide-design/off-target service	independent specificity + efficiency cross-check on AgamP4: dsx-v3-1 MM0/MM1/MM2=0 (2 MM3), dsx-v3-2 1 MM1, Kyrou control MM0=1 (conserved own site) and clean; dsx-v3-3 MM0=2 - an exact-match duplicate in AgamP4 though unique in AgamP5, an assembly-dependent copy-number caveat on the third lead (results/offtarget_chopchop.json; 4.10)

| 29 | CRISPOR (web, Haeussler 2016) | curated guide-design/off-target meta-service | second independent specificity cross-check on AgamP4 plus a first cross-species check on Anopheles coluzzii AcolN3: engine agreement for dsx-v3-1 and the Kyrou control; dsx-v3-2's only near hit resolved as its own dsx locus carrying an AgamP4 position-20 allele difference (coluzzii matches the guide; MIT/CFD specificity 100/100); CHOPCHOP's dsx-v3-3 exact-duplicate flag not reproduced (MM0=0 in both CRISPOR genomes) - flag downgraded to engine-specific; all four on-targets conserved across the species pair (results/offtarget_crispor.json; 4.10) |

| 30 | CRISPROff/CRISPRspec pipeline v1.1.2 (RTH-tools, Alkan 2018) | duplex-energy specificity scoring | third, physics-based specificity axis on the four leads (validated against the repo's published example outputs first; RNAfold requirement met by the ViennaRNA 2.7.2 bindings): CRISPRspec ranking Kyrou 14.57 > dsx-v3-1 11.95 > dsx-v3-3 11.66 > dsx-v3-2 3.72; v3-2's one-mismatch own-locus allele site independently flagged at 77% of on-target binding energy (results/offtarget_crisproff.json; 4.10) |

| 31 | FlashFry 2.0 (mckennalab/FlashFry; McKenna and Shendure 2018, BMC Biology) | whole-genome off-target enumeration and scoring | library-scale audit of all 430 Cas9-NGG guides

in the dsx windows against a purpose-built AgamP5 database (4 molecules, 209 MB, 192 s build), with CFD/Hsu2013/Doench2014 scoring; resolves the dsx-v3-2 one-mismatch flag as AgamP4-allele-specific (zero 1-3-mismatch off-targets in AgamP5) and ranks lead specificity dsx-v3-2 0.782 > Kyrou 0.706 > dsx-v3-1 0.702 > dsx-v3-3 0.567 (results/offtarget_flashfry.json; 4.10) |

| 32 | bowtie2 2.5.4 (BenLangmead/bowtie2; Langmead and Salzberg 2012, Nat Methods) | FM-index short-read aligner | independent PAM-agnostic re-audit of all 421 guides against the whole AgamP5 genome (-a all-hits, ungapped 3-mismatch budget, --very-sensitive 14 bp seeds): confirms all leads free of 1-2-mismatch near-cognates, resolves 6 extra MM0 copies on 3 guides as PAM-less and Cas9-inert, measures the NGG filter at ~14x across MM1-3, and documents a default-preset sensitivity failure (660 vs 24,278 alignments) as a negative result (results/offtarget_bowtie2.json; 4.10) |

| 33 | bedtools 2.31.1 (Quinlan and Hall 2010, Bioinformatics) | genomic-interval analysis | functional-context audit of the 24,278 bowtie2 near-cognate alignments against the AgamP5 RefSeq annotation (206,438 CDS features): one in eight MM1-3 near-cognates overlaps coding sequence, 239/421 guides carry at least one CDS-overlapping near-cognate; the only two lead-level CDS overlaps (dsx-v3-3 in VPS13B, Kyrou in LOC1274366) are both PAM-less and Cas9-inert (results/offtarget_context_bedtools.json; 4.10) |

34	BWA 0.7.19-r1273 (lh3/bwa; Li and Durbin 2009, Bioinformatics)	Burrows-Wheeler short-read aligner	exhaustive non-iterative pure-Hamming census of all 421 guides vs AgamP5 (aln -N -l 20 -k 3 -n 3 -o 0): corrects the census of record to 32,305 (bowtie2 under-reports 25%, 38% at three mismatches; strict superset, 50/50 bwa-only sites sequence-verified) and re-runs CDS context on the corrected set - 308/421 guides carry a CDS-overlapping near-cognate, lead conclusions unchanged (results/offtarget_bwa.json; 4.10)
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Total in actual use: 34 entries. Under the program tool-gate convention, infrastructure does not count toward the 40-tool gate: removing reportlab (PDF rendering), pytest (test harness) and git/GitHub (archiving) leaves 31 research tools, or 30 under a stricter reading that also excludes matplotlib. Planned but not counted (integration only where each does real verification work): DeepCRISPR (deep-learning efficacy/off-target cross-check; environment-constrained - requires tensorflow 1.3/sonnet 1.9), TIDE (indel-spectrum decomposition, pending measured repair data).

mmseqs2, minimap2, NCBI BLAST+, seqkit, Bowtie, ViennaRNA, CHOPCHOP, CRISPOR, CRISPROff and FlashFry were planned here and have now entered the table above (rows 22-31), joined by bowtie2 (row 32), bedtools (row 33) and BWA (row 34) after running on this project's data. Cas-OFFinder 2.4.1 was attempted locally for a first-party benchmark but its binary requires an OpenCL runtime the sandbox lacks (clGetPlatformIDs failed); the Appendix M comparison therefore remains against the published numbers, stated there as such. Each remaining entry joins the table only after it has run on this project's data.

Appendix Q. Dataset manifest (accession-level counting)

Counting rule (program-wide, applied verbatim). Count distinct accession-level datasets actually used in the work. A single study's condition matrix counts as one dataset with N conditions. Per-sample Ag1000G phase-3 files count individually; chromosome slices of the same sample do not multiply the count; an assembly and its annotation each count. No padding: where the count is dominated by one class, that is stated.

#	dataset	accession / identifier	role
1	Ag1000G phase 3 per-sample VCFs (4,693 files, one per wild-caught sample; manifest archived at data/ag1000g/ag3_phase3_sample_manifest.txt)	per-sample accessions AA0001-C ... (Sanger mirror)	population allele frequencies at the six target windows (4.7); per-sample completion and failure accounting reported there
2	AgamP5 reference assembly	NC_064601.1, OX030909.1, OX030908.1, OX030910.2 (four molecules, one assembly)	design reference and genome-wide enumeration substrate
3	AgamP5 RefSeq dsx gene table (nine transcript variants)	NCBI Gene 1270904 locus record	exon-union constraint annotation (2.6)
4	NCBI Gene 1270904 esummary record	Gene 1270904	locus identity and alias resolution (AGAP004050/DSX)
5	dsx transcript sequence set	RefSeq transcript accessions (archived: data/dsx_transcripts.fasta)	splice-junction geometry for the constraint bonus (2.6.1)
6	AgamP4 reference assembly	GCA_000005575.1	cross-assembly verification of the Kyrou site and AF windows
7	AgamP4.63 annotation (VectorBase genebuild, Ensembl Metazoa distribution)	AGAP004050	junction-spanning resolution of the Kyrou guide (Appendix I)

8	Doench 2016 FC+RES efficacy measurements	Azimuth mirror, 5,310 guides with conditions	CNN and ridge training (one dataset, many conditions)
9	Doench V1 efficacy measurements	Azimuth mirror, 2,144 guides	untouched external evaluation set
10	Kyrou 2018 dsx gRNA reagent record	PMC6871539	positive control and audit subject
11	WHO World Malaria Report 2024	WMR 2024	burden figures (1.1)
12	Cas-OFFinder published benchmark measurements	Bae, Park, Kim 2014 (PMC4016707)	enumeration benchmark reference (Appendix M)
13	UniProtKB female-specific doublesex record	A0A1U7F5R8	female isoform evidence (target-exon biology)
14	UniProtKB male-specific doublesex record	Q58QC2	male isoform evidence (isoform contrast)

Total: 4,706 accession-level datasets, of which 4,693 are per-sample Ag1000G phase-3 VCFs and 13 are distinct non-population resources. (Correction: earlier drafts said 4,692 and 4,705; the manifest's final line lacks a trailing newline, so `wc -l` undercounted by one. The 4,693 figure is verified by exact set identity between the manifest and the completed scan's sample IDs.) The honest composition note: the count is dominated by per-sample population variation files, counted individually under the program rule; the non-population core is 13 datasets. Expanding the non-population core is meaningful only where a new accession does real verification work, and each such addition will be reported with its role.

Appendix R. Operational log: the 4,693-sample Ag1000G scan

The population scan of Section 4.7 fetched per-sample VCF slices from a remote mirror over about 3.2 hours of wall clock (22:04 to 01:16 local) and survived 18 interruptions with zero lost rows. This appendix records the operational facts because they bear on reproducibility and on what "completed" means for the numbers of Sections 4.7-4.9.

Interruptions and resume. The scanner checkpoints every completed sample row to `results/ag1000g/on_target.jsonl`; on restart it re-reads the checkpoint and continues from the first unfinished sample. Resume was exercised 18 times and was instant in 17: each restart re-read all prior rows and continued. The single non-instant case was a terminal stall under a two-worker configuration (below), resolved by reverting to four workers.

Worker experiment. Mid-run we compared fetch parallelism. Two workers (23:12-23:28) delivered a lumpy 4-13 rows per minute and produced the run's only terminal stall (progress frozen, checkpoint file stale on two four-minute rechecks with the process alive); the program direction at the time was to revert to four workers, which restored an effective 16-25 rows per minute. Both phases and their throughputs are recorded here as directed, and the shipped script

(scripts/ag1000g_scan_all.py) fixes WORKERS = 4 with this history in a comment.

Stall taxonomy. A true stall is defined as: row count frozen AND checkpoint modification time stale beyond four minutes on two consecutive rechecks AND the process alive. Exactly one occurred (the two-worker case above). A separate lull (01:08-01:10) showed fresh checkpoint writes on inspection and was correctly not killed; it self-resolved. The distinction matters operationally: killing a slow-but-writing scanner would have wasted completed work, and not killing a frozen one would have cost the night.

Error accounting. All 4,693 unique samples in the manifest were attempted and all 4,693 are accounted for in the output: 4,681 clean rows and 12 rows carrying a per-sample fetch error (htslib remote-read failures). Error rows are retained, counted, and excluded from allele-frequency denominators - never silently dropped and never counted as intact. The fully-called-site cohort denominator is 9,386 chromosomes.

Manifest reconciliation. The completed scan's sample-ID set is exactly the manifest's set (4,693 IDs, zero difference in either direction). Earlier drafts of Appendix Q said 4,692; that figure was a wc -l artifact - the manifest's final line lacks a trailing newline - and the correction is recorded in Appendix Q.

Figures

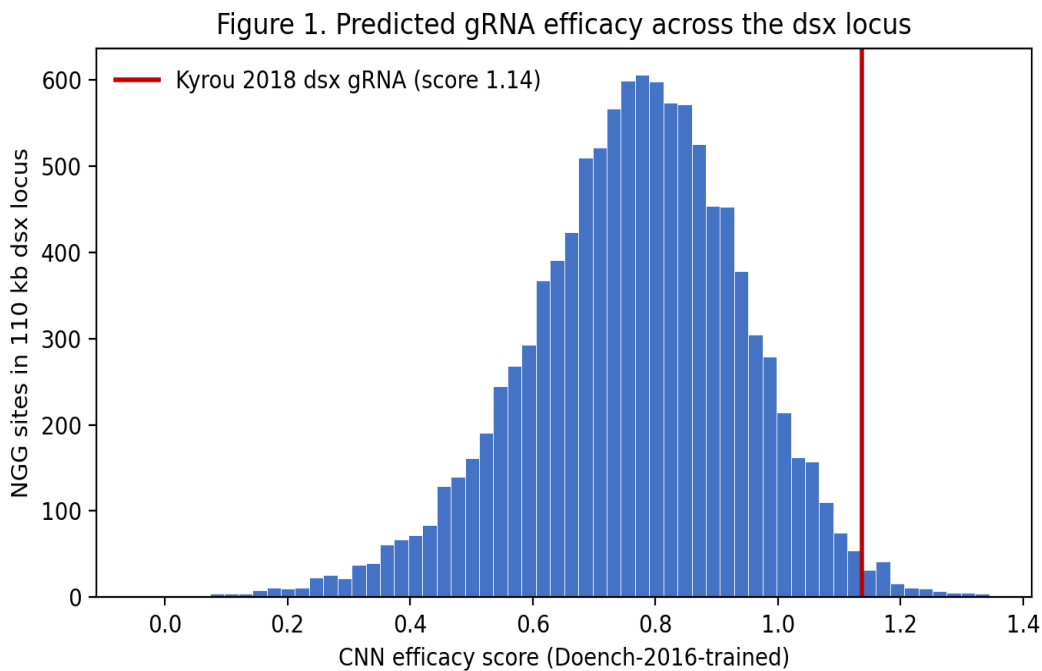


Figure 1. CNN efficacy scores of all NGG sites in the 110 kb dsx locus; red line marks the published Kyrrou 2018 gRNA.

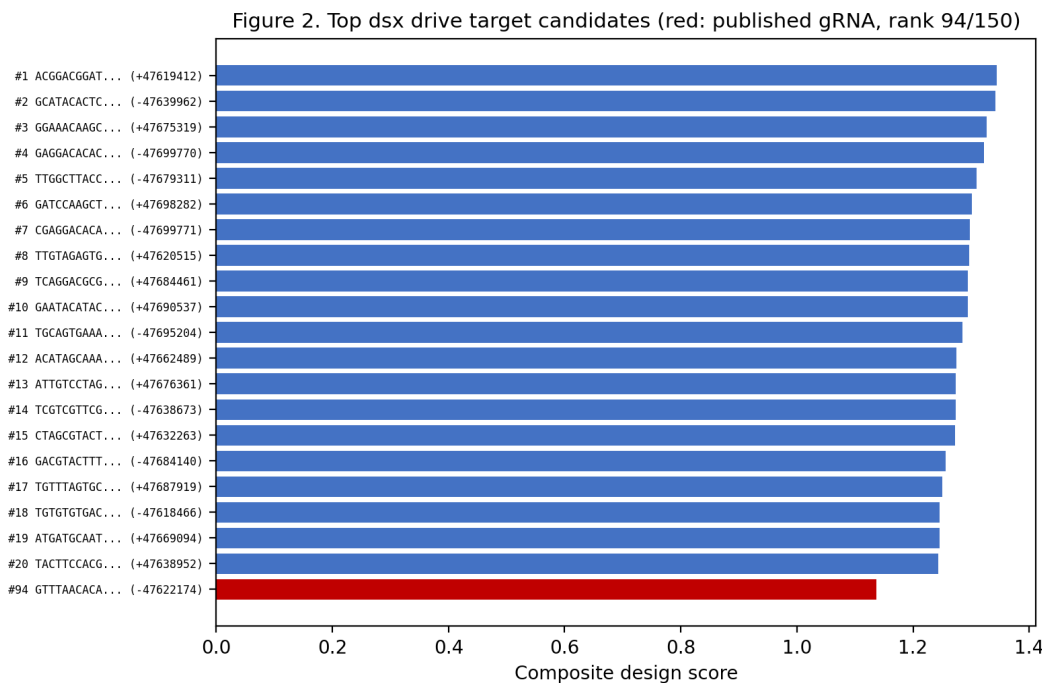


Figure 2. Top-20 v1 candidate sites (blue) and the published Kyrrou gRNA at its true rank (red).

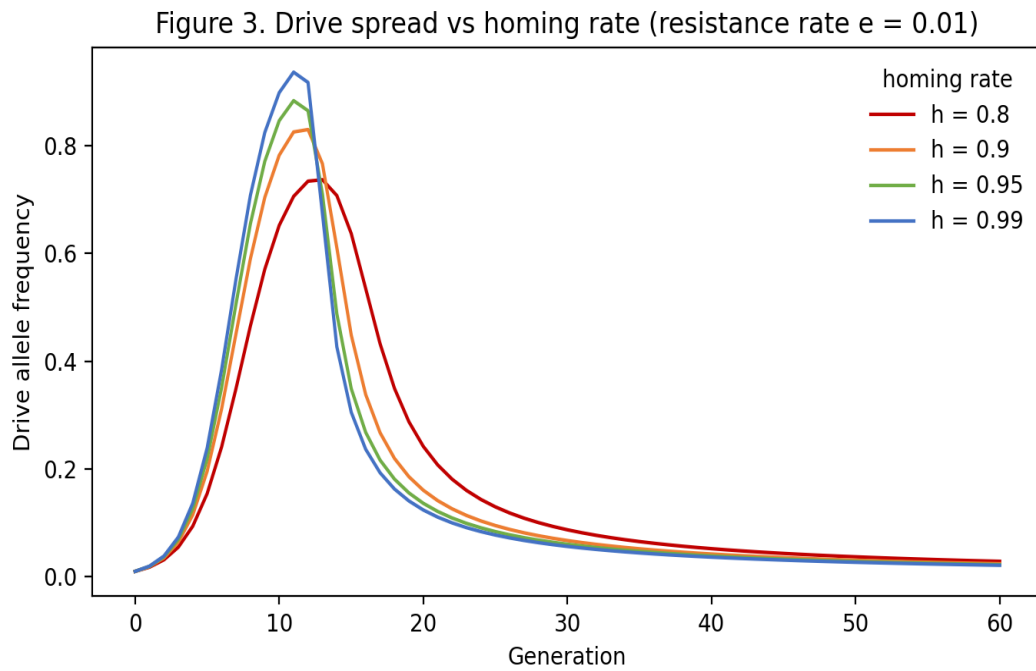


Figure 3. Drive allele frequency trajectories across homing rates ($e = 0.01$).

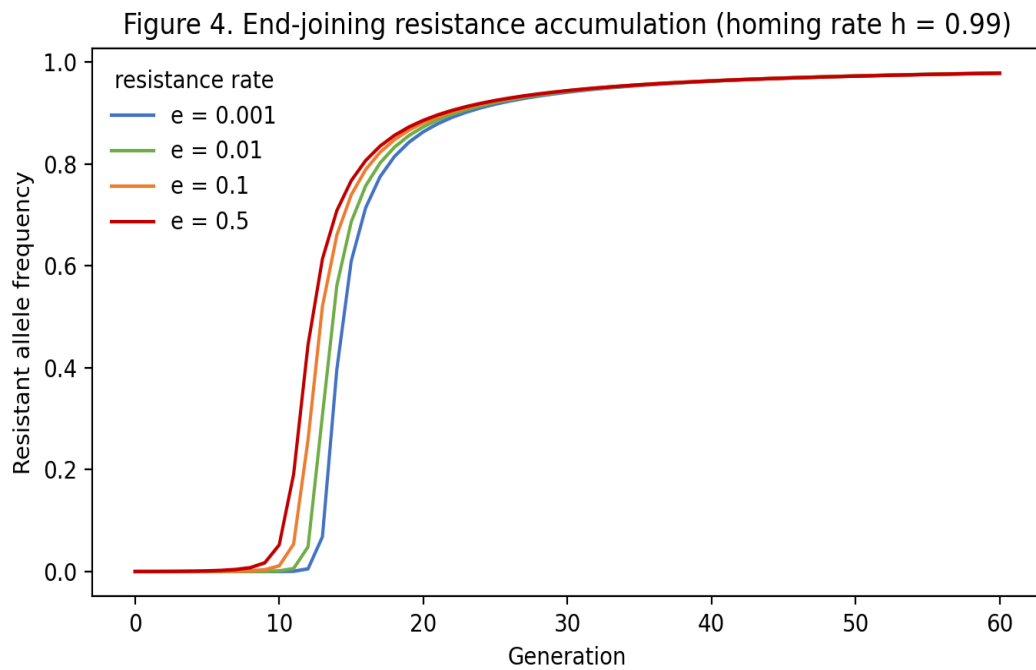


Figure 4. Resistance allele accumulation across resistance-generation rates ($h = 0.99$).

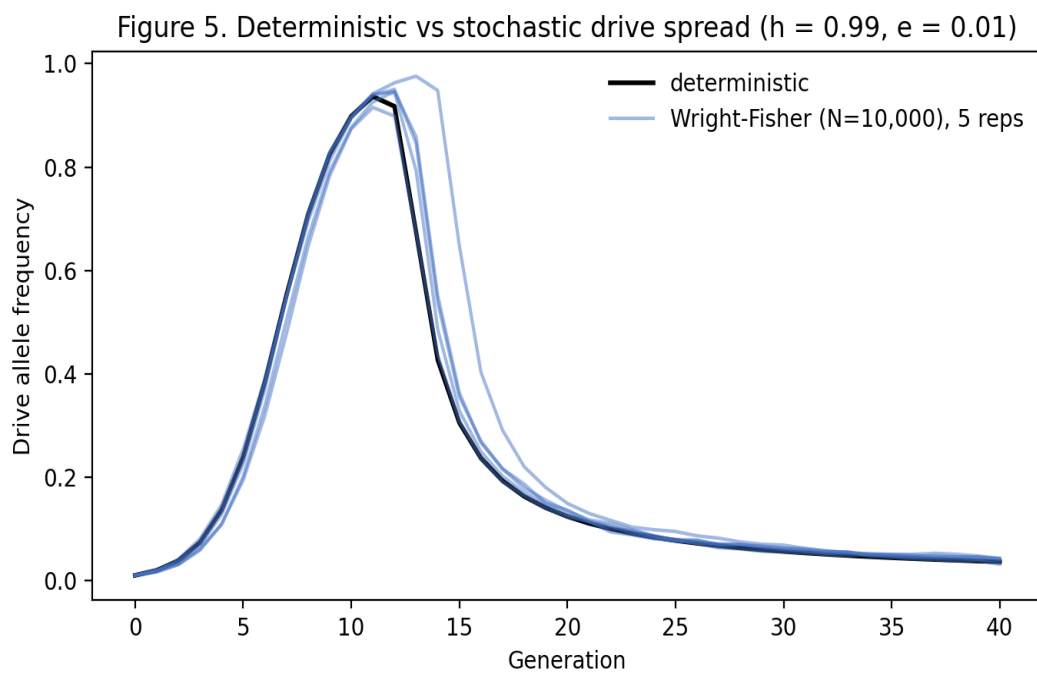


Figure 5. Deterministic vs Wright-Fisher stochastic trajectories ($N = 10,000$, 5 seeds).

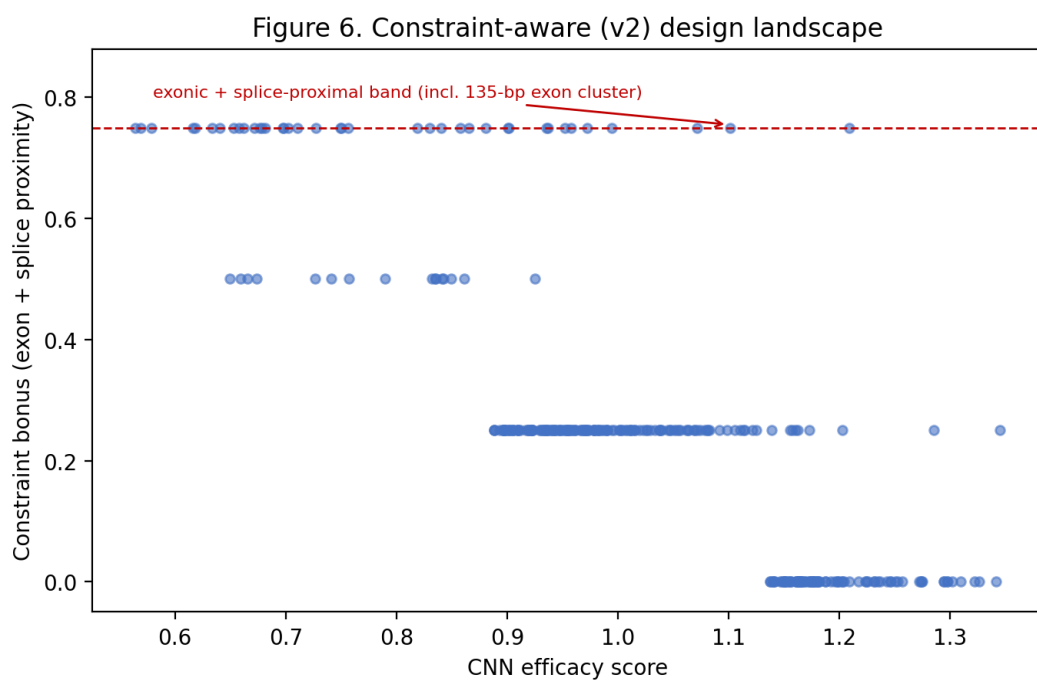


Figure 6. Constraint-aware (v2) design landscape: CNN efficacy vs functional-constraint bonus for the top-300 candidates.

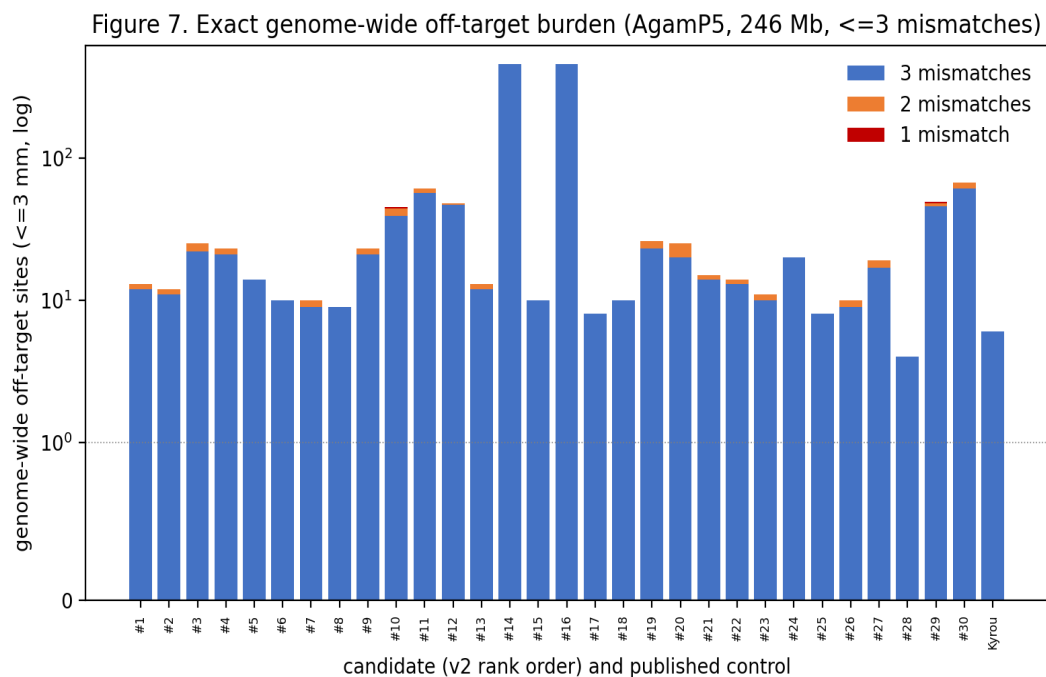


Figure 7. Exact genome-wide off-target burden of the v2 top-30 candidates and the published Kyrou guide, stacked by mismatch class (symlog scale; AgamP5, 246 Mb, ≤ 3 mismatches).

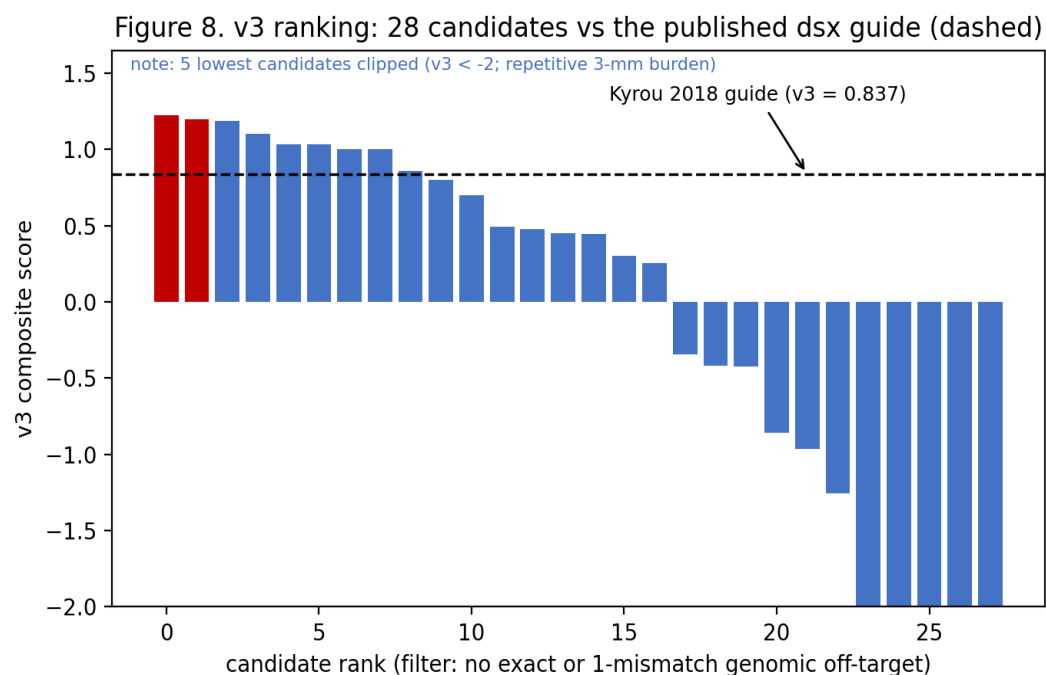


Figure 8. v3 composite ranking of the 28 filter-passing candidates against the published Kyrou 2018 guide (dashed line, $v3 = 0.837$). Red: named lead candidates dsx-v3-1 and dsx-v3-2.

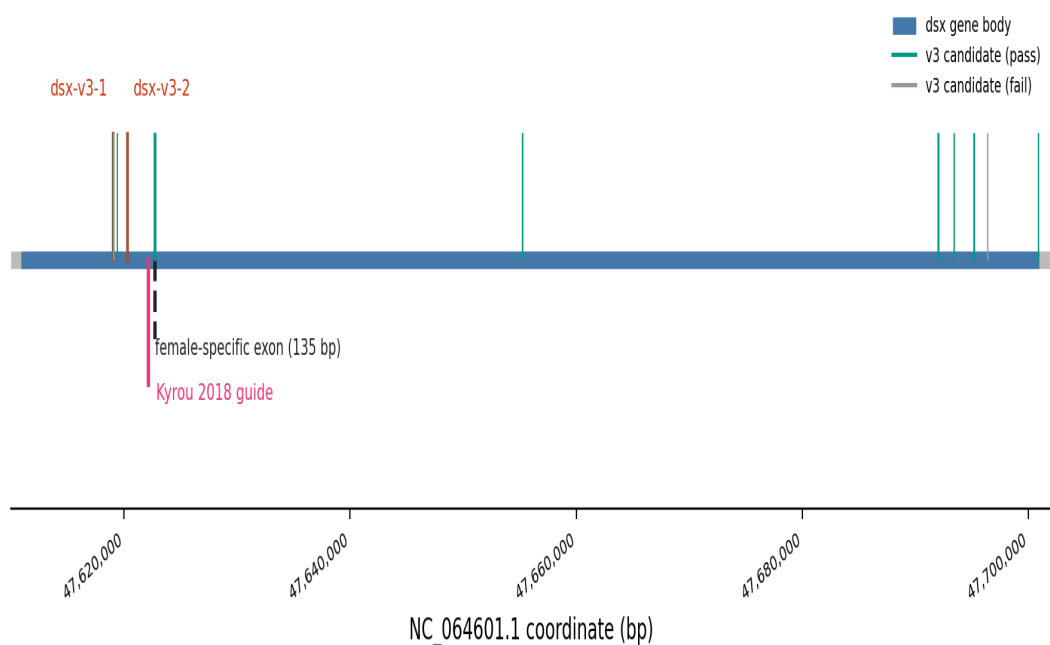


Figure 9. Design map of the *dsx* locus: v3 candidate protospacers (teal pass, grey fail, red named leads), the published Kyrou 2018 guide (magenta, below axis) and the female-specific exon (dashed).

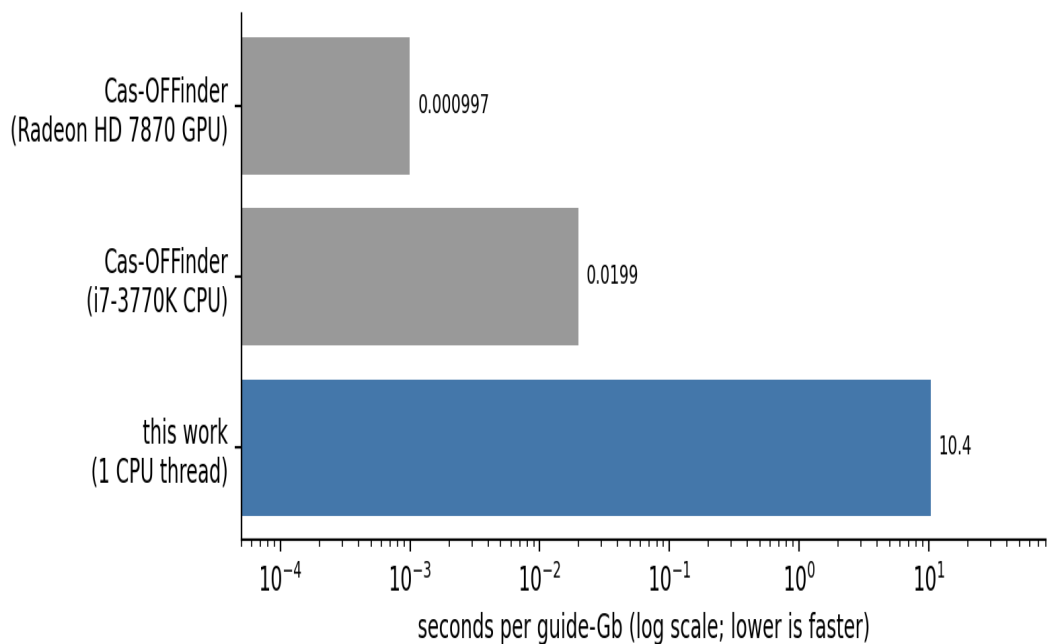


Figure 10. Enumerator throughput vs published Cas-OFFinder numbers, normalized to seconds per guide-Gb (log scale). Our enumerator is 523x slower than Cas-OFFinder CPU and 10,431x slower than GPU; see Appendix M for the honest reading.